

NETWORK FORENSIC INVESTIGATION REPORT

DIGITAL FORENSIC INVESTIGATION REPORT

Nitroba University Harassment Case — Digital Evidence Examination

Case Title	Nitroba University Harassment Case — Digital Evidence Examination
Case Identifier	CIP-B105-CS2-C11/26/DFIT/17310
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Report Status	DRAFT (Working Merged Analysis Report)

STUDENT DECLARATION

Declaration of Original Work

I hereby declare that this forensic investigation report, the accompanying master provenance calculations, evidence correlation matrices, and analytical conclusions are entirely my own original work.

All derived sensitive image evidence and network extraction artifacts generated during this assessment have been handled strictly inside the isolated, protected IC DFA laboratory environment in compliance with safety controls. I confirm that original source evidence files have remained unedited, and all temporary working derivatives will undergo secure, authorized deletion (shredding) following final evaluation in accordance with IC DFA retention policy.

Field	Value
Student Name	Ebenezer Elikem Amankwah Ofori Dzam
Date	September 15, 2026
Signature	E. E. A. O. Dzam

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Executive Summary

Lily Tuckrige, a Chemistry 109 teacher, reported harassing e-mail sent to her personal Yahoo account. The originating IP of the first message pointed only to a shared student-accommodation address, which cannot by itself identify an individual. A later message referenced a self-destructing-message service, willselfdestruct.com. A network capture taken on the dorm Ethernet uplink (nitroba.pcap) was examined to establish whether a specific device, and from that device a specific student on the Chemistry 109 roster, can be tied to that submission.

The examination recovered the harassing message in full clear text from an HTTP POST request, tied that request to a specific client device by MAC address, correlated that device with an e-mail-bearing identity artefact (session cookies) recurring across 149 frames and two independent web services, and cross-checked that identity against the supplied class roster. Every step is anchored to a specific, reproducible packet or filter query rather than to inference from the original complaint alone.

Key finding: The device at MAC 00:17:f2:e2:c0:ce submitted the harassing message to willselfdestruct.com at 2008-07-22T06:04:24Z, addressed to the victim's reported address. The same device is strongly associated with the account jcoachj@gmail.com, which corresponds uniquely to Johnny Coach on the Chemistry 109 roster.

Overall confidence: HIGH for device/account attribution; MODERATE-HIGH for final human identification, given the open/shared nature of the dorm network (see Q8).

This report is organised so that authority and scope are stated before any finding, every finding is supported by a cited artefact and reproducible query, and limitations are stated explicitly rather than implied.

1. Authority, Scope and Investigative Questions

1.1 Authority

This examination was conducted as a self-paced module exercise using a supplied, publicly released historical training capture (the 2008 Nitroba University corpus, distributed by digitalcorpora.org for digital-forensics education). No live systems, real individuals, or non-consenting third parties were accessed. The case package's own "Authorisation and Evidence Safety" instructions were treated as binding throughout: use only the supplied historical capture; do not contact the individuals named in the scenario or submit messages to historical services; do not alter the original PCAP; do not use MAC-address information to track real devices outside the supplied case; redact unnecessarily explicit or personally identifying content from any screenshots produced.

1.2 Scope

In scope: the supplied nitroba.pcap capture in its entirety; comparison of any recovered identity artefact against the supplied Chemistry 109 class roster; documentation of tools, filters and commands sufficient for another analyst to reproduce every finding.

Out of scope: any network, host, or account not represented in the supplied capture; verification of account ownership through any channel outside the packet evidence itself (e.g. contacting the platform, the university, or the named individuals); assessment of criminal or disciplinary liability, which is a matter for the institution's own process and outside a technical attribution report.

1.3 Investigative Questions

The following questions were set by the case brief (“Required Findings”) and are answered in turn in §6:

- Q1. Which IP accessed willselfdestruct.com?
- Q2. Does the relevant IP submit data to the website?
- Q3. Does the POST contain the harassment message or fields consistent with it?
- Q4. Which MAC address is associated with the relevant client traffic?
- Q5. What identity-bearing cookie/email artefacts are associated with that MAC?
- Q6. Is the attributed identity in Chemistry 109?
- Q7. When was the service used?
- Q8. What packet/stream evidence supports each attribution step?
- Q9. What alternative explanations remain because the dorm network was shared/open?
- Q10. What is the final attribution confidence?

2. Evidence Inventory and Integrity Verification

A full, itemised evidence register with hashes, sources and exhibit references is at Appendix B; this section summarises the primary evidence item and the integrity procedure applied to it.

Property	Value
Exhibit	EV-01 — nitroba.pcap
Size	56,180,821 bytes
MD5	9981827f11968773ff815e39f5458ec8
Source	https://downloads.digitalcorpora.org/corpora/network-package-dumps/2008-nitroba/nitroba.pcap
Acquisition method	wget (direct HTTPS download from the public digitalcorpora.org corpus)
Hash computed	openssl dgst -md5 nitroba.pcap — immediately after download, before any filtering
Chain of custody	Single analyst, single working copy; original file not modified (read-only Wireshark/TShark display-filter analysis only — no capture-file edits, no re-saves over the original)
Total frames / duration	94,410 frames / 15,760.0 s (4h 22m 40s)

The hash was recomputed at a later point in the session (Appendix A, 15:48) and matched the value recorded at acquisition, confirming the working file was not altered between the start and end of analysis.

3. Tools, Versions and Methodology

3.1 Tools and environment

Tool	Role in this examination
Kali Linux	Analysis workstation OS
Wireshark	Interactive packet inspection, protocol decoding, Follow-TCP-Stream, packet-detail review
TShark	Repeatable command-line filtering, field extraction, IO statistics
OpenSSL	MD5 hashing of the evidence file for integrity verification

Exact tool build/version numbers were not visible in the analysis-session screenshots supplied for this report. Before any live reproduction or defense, the analyst should capture and record wireshark --version, tshark --version and openssl version output for the record.

3.2 Methodology

Analysis combined Wireshark (interactive inspection, packet-detail decoding) and TShark (repeatable command-line filtering and field extraction), so that every filter used to reach a finding can be re-run by another analyst against the same hash-verified capture. The general sequence followed was: (1) preserve and hash the evidence; (2) establish capture scope, duration and endpoints; (3) reconstruct the site-access/session flow for the website named in the complaint; (4) isolate and decode the HTTP POST; (5) identify the client device by MAC address and quantify its activity; (6) search that device's traffic for identity-bearing artefacts; (7) normalise all timestamps to UTC and build an incident timeline; (8) compare the recovered identity against the supplied roster; (9) assemble a packet-level attribution matrix; (10) state limitations and a confidence-rated conclusion.

3.3 Reproducibility

The full filter/command log is reproduced as Appendix D. Every finding in §6 cites the specific frame number(s) and the exact TShark filter or Wireshark display filter used to reach it, so the finding can be independently reproduced from the hash-verified EV-01 working copy.

4. Detailed Findings and Evidential Interpretation

Findings are presented in the order of the investigative questions set in §1.3. Each finding states the artefact, its exact locator, the interpretation, and — where relevant — what the artefact does not prove.

4.1 Q1 — Which IP accessed willselfdestruct.com?

Finding: 192.168.15.4.

Locator: DNS resolution ($\approx$ frame 82923: `www.willselfdestruct.com` → CNAME `willselfdestruct.com` → A `69.25.94.22`) followed by TCP/HTTP traffic from 192.168.15.4 to 69.25.94.22, e.g. frame 82936.

Interpretation: 192.168.15.4 is the only client IP observed initiating the DNS lookup and subsequent HTTP session with the resolved `willselfdestruct.com` address in this capture window.

4.2 Q2 — Does the relevant IP submit data to the website?

Finding: Yes.

Locator: Frame 83601 — filter `(ip.src==192.168.15.4) && (ip.dst==69.25.94.22) && http.request.method=="POST"`.

Interpretation: 192.168.15.4 issued a single HTTP POST to `/secure/submit`, Content-Type `application/x-www-form-urlencoded`, 39.7 seconds after loading the submission form — consistent with a human filling in and submitting the on-screen form rather than automated/background traffic.

4.3 Q3 — Does the POST contain the harassment message or fields consistent with it?

Finding: Yes.

Locator: Frame 83601, decoded HTML form fields (`tshark -Y "frame.number==83601" -V | grep -i "form item"`).

Form field	Recovered value
to	lilytuckrige@yahoo.com
from	(empty)
subject	you can't find us
message	and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running.
type / ttl	0 / 30
submit.x / submit.y	92 / 26 (image-button click coordinates — confirms an interactive browser submit)

Interpretation: the “to” address is an exact match for the victim's reported Yahoo address, and the message text (“Stop teaching. Start running.”) is directly consistent with the reported harassment. This is direct, recovered evidence, not inference.

4.4 Q4 — Which MAC address is associated with the relevant client traffic?

Finding: 00:17:f2:e2:c0:ce (OUI-resolved to Apple, Inc.).

Locator: Ethernet II source address on frame 83601 (tshark -Y "frame.number==83601" -T fields -e eth.src).

Corroborating volume evidence: eth.addr==00:17:f2:e2:c0:ce matches 73,246 of 94,410 frames (77.6%) and 44,742,268 bytes across the full 4h22m capture — the profile of one continuously active personal device rather than a shared or momentarily-used one. The destination MAC on the same frame, 00:1d:d9:2e:4f:60 (“HonHaiPrecis”), is consistent with the dorm segment's router/gateway, not the suspect device.

4.5 Q5 — What identity-bearing cookie/email artefacts are associated with that MAC?

Finding: jcoachj@gmail.com, recurring in two independent cookie mechanisms.

Locator: eth.src==00:17:f2:e2:c0:ce && http.cookie contains "@" → 149 matching frames.

Cookie	Value	Service context
gmailchat	jcoachj@gmail.com/475090	Gmail chat-presence cookie (mail.google.com)
OL_SESSION	jcoachj@gmail.com-cal	Outlook/Live calendar session cookie

Interpretation: two unrelated web services independently reflect the same account identifier on the same client MAC, which is materially stronger corroboration than a single cookie sighting.

4.6 Q6 — Is the attributed identity in Chemistry 109?

Finding: Yes — Johnny Coach.

Locator: Supplied “Chem 109 — Class roster” slide, compared against jcoachj@gmail.com under a first-initial + surname (+ trailing initial) convention.

Johnny Coach is the only “Coach” on the roster and the only surname from the roster that appears in the recovered address under any reasonable construction. This is a naming-pattern correlation and is treated as circumstantial, not conclusive, evidence (see Q7).

4.7 Q7 — When was the service used?

Finding: Form loaded 2008-07-22T06:03:43.83Z; harassing message posted 2008-07-22T06:04:24.31Z (UTC). Full detail at 5.

4.8 Q8 — What packet/stream evidence supports each attribution step?

Consolidated in the Packet-Level Attribution Matrix, Table 4.8.1.

Layer	Value	Source artefact / locator	Strength
Complaint IP (first e-mail)	Shared student-accommodation IP (per victim report)	Case narrative / e-mail header (outside this capture)	Weak alone — shared IP
Website / service IP	69.25.94.22 (willselfdestruct.com)	DNS response ~frame 82923; HTTP frames 82936–83654	Confirmed
Client IP	192.168.15.4	All GET/POST frames to 69.25.94.22, e.g. 82936, 83601	Confirmed for this session
HTTP POST content	to=lilytuckrige@yahoo.com; “Stop teaching. Start running.”	Frame 83601 (§4.3)	Confirmed
Client MAC	00:17:f2:e2:c0:ce (Apple)	eth.src, frame 83601 (§4.4)	Confirmed for the POST frame
Identity artefact	jcoachj@gmail.com	149 frames from the same MAC (§4.5)	Strong, cross-service
Roster match	Johnny Coach	Chem 109 roster (§4.6)	Circumstantial

4.9 Q9 — Alternative explanations (shared/open network)

Addressed fully in 7; summary: IP alone was never relied upon for attribution; MAC identifies a device, not conclusively a person; cookies show authenticated use, not exclusive possession.

4.10 Q10 — Final attribution confidence

HIGH for device/account attribution; MODERATE-HIGH for final human identification. Full reasoning in 7 and 8.

5. Integrated Timeline

Two distinct clocks are in evidence in this case: the UTC-normalised packet timeline (when the incident itself occurred, in 2008) and the analysis workstation's session clock (when the analyst performed each examination step). These are kept as two clearly labelled sub-timelines rather than merged into one sequence, because the workstation clock recorded only hour:minute with no date or timezone metadata and cannot be safely correlated to the second against the 2008 UTC evidence — conflating them would overstate precision. The full investigative activity log with source screenshots is at Appendix A.

5.1 Incident timeline (UTC, from packet evidence)

UTC time	Event	Locator
2008-07-22T01:51:07.095278Z	Packet capture begins	Frame 1
2008-07-22T06:03:43.825871Z	Client 192.168.15.4 requests the willselfdestruct.com submission form	Frame 82936
2008-07-22T06:03:44.210946Z	Server returns the form; issues JSESSIONID cookie	Frame 82998
2008-07-22T06:04:24.311700Z	Client submits the harassing message via POST /secure/submit	Frame 83601
2008-07-22T06:04:24.564165Z	Client requests /secure/success — server confirms receipt	Frame 83614
2008-07-22T06:04:24.902157Z	Confirmation page renders (bk-message.gif)	Frame 83654

UTC time	Event	Locator
2008-07-22T06:13:47.046029Z	Packet capture ends	Frame 94,410

Timezone note: TShark reported every frame's local time as UTC-05:00 (US Central Daylight Time), which was used as the sole, documented basis for UTC conversion throughout this report. This is distinct from, and not to be confused with, the willselfdestruct.com server's own HTTP "Date" response header ("Tue, 22 Jul 2008 07:24:05 GMT" on frame 82998), which is offset from the capture-derived time by roughly 1h20m — the remote server's clock is evidently not synchronised with the capture host and was deliberately not used for sequencing events.

5.2 Investigative activity timeline (analysis session)

Summarised here; full detail (all 26 session screenshots, cross-referenced) is at Appendix A.

Session time	Activity
14:42	Preserved the assigned evidence file and computed a cryptographic hash prior to any analysis: <code>openssl dgst -md5 nitroba.pcap</code> .
14:46	Opened the hashed working copy in Wireshark; performed an initial scan of the end of the capture to confirm scope, protocols and endpoint activity.
14:52	Applied filter <code>http.request.method=="GET" && http.host=="www.willselfdestruct.com"</code> ; located the client's GET /secure/submit request (frame 82936).
14:55	Reviewed the corresponding HTTP 200 OK response (frame 82998), noting the Set-Cookie: JSESSIONID value and Apache server banner.
15:01	Applied filter <code>(ip.src==192.168.15.4) && (ip.dst==69.25.94.22) && http.request.method=="POST"</code> ; located and decoded frame 83601, recovering the submitted form fields.
15:04	Expanded the Ethernet II layer of frame 83601 to record source/destination hardware addresses.
15:08	Applied filter <code>eth.addr==00:17:f2:e2:c0:ce</code> to assess the proportion of total capture traffic associated with the client MAC.
15:11	Applied filter <code>eth.src==00:17:f2:e2:c0:ce && http.cookie contains "@"</code> ; identified a Gmail chat-presence cookie carrying an e-mail-derived identifier.
15:18	Set Wireshark's time display format to UTC Date and Time of Day to prepare for timestamp normalisation.
15:23	Located the POST's corresponding server confirmation response containing the phrase "secure anonymous E-mail".
15:48	(TShark, terminal) Re-confirmed acquisition provenance (<code>digitalcorpora.org 2008-nitroba corpus</code>) and re-verified the MD5 hash of the working file.
15:50	<code>tshark -r nitroba.pcap head/tail</code> — confirmed first and last frame numbers and summary content.
15:51	<code>tshark -qz io,stat</code> — confirmed total frame count (94,410) and capture duration (15,760.0 s).
15:52	<code>tshark -T fields -e frame.time -e frame head</code> — recorded the absolute local timestamp of frame 1 (capture start).
15:54	<code>tshark -Y (GET filter)</code> — produced the full chronological sequence of requests to <code>www.willselfdestruct.com</code> .
15:55	Re-ran the GET-sequence query with <code>-T fields (frame.time, frame.time_delta_displayed, ip.src, ip.dst, http.request.uri)</code> for an exact, exportable timeline.
15:56	<code>tshark -Y "data-text-lines matches..."</code> — isolated the two HTTP responses containing the confirmation-page text.
16:03	<code>tshark -Y (POST filter) and frame 83601 -V grep</code> — cross-verified the POST content and confirmed the "to" field against the victim's reported address.

Session time	Activity
16:05	tshark -T fields -e frame.time -e text — extracted the POST's exact timestamp together with the full decoded form-field text in one query.
16:06	Re-ran frame 83601 -V grep "form item" for a clean, isolated listing of all nine submitted form fields.
16:09	Confirmed the urlencoded-form.key / urlencoded-form.value TShark field definitions used for the structured extraction.
16:13	Confirmed eth.src for frame 83601 and quantified total frames/bytes associated with that MAC via io.stat.
16:15	Counted frames from the client MAC containing an "@"-bearing cookie value (149 frames).
16:16	Ran a regex extraction (-V grep -P) across all matching frames to enumerate every recovered e-mail-pattern cookie value from the client MAC.
16:17	Re-confirmed the exact POST frame timestamp for cross-reference in the integrated timeline.
—	Reviewed the supplied Chemistry 109 class-roster slide for comparison against the recovered identity artefact.

6. Alternative Explanations, Limitations and Confidence Assessment

6.1 Alternative explanations considered

- Shared/open network: the dorm segment carries traffic for multiple occupants behind a common uplink; IP address alone (as used for the original complaint e-mail) cannot identify an individual and was deliberately not relied on here beyond providing the initial lead.
- Device ≠ person: MAC-address attribution identifies a network interface, not conclusively the person using it at 06:04 UTC. Roommates, guests, or a borrowed/compromised device remain a theoretical alternative not excluded by packet evidence alone.
- Cookie ≠ exclusive ownership: authenticated-session cookies show that an account (jcoachj@gmail.com) was logged in on this device across many requests, which is strong usage evidence but not cryptographic proof that no other individual ever used the same machine.
- Roster match is a name pattern, not a verified account registry lookup: “jcoachj” corresponding to “Johnny Coach” is inferred from a common naming convention, not confirmed against an authoritative account-ownership record.
- Capture-scope limitation: the sniffer captured only the dorm Ethernet uplink; it cannot speak to activity on devices that never traversed that link, and cannot independently verify who was physically at the keyboard.
- Server clock skew: the willselfdestruct.com server's own HTTP Date header is skewed by roughly 1h20m relative to the capture host's clock — a reminder that timestamp evidence must be sourced from one authoritative, documented clock rather than mixed uncritically across systems.

6.2 Confidence assessment by attribution layer

Attribution layer	Confidence	Basis
Website/service identification	HIGH	Direct DNS + HTTP evidence
POST content = harassing message	HIGH	Recovered verbatim, exact victim-address match
Client MAC for the POST	HIGH	Direct eth.src on the POST frame
Identity artefact (jcoachj@gmail.com)	HIGH (device/account level)	Two independent services, 149 corroborating frames

Attribution layer	Confidence	Basis
Roster match (Johnny Coach)	MODERATE	Naming-pattern inference, not a verified registry lookup
Final human attribution	MODERATE-HIGH	Open/shared network means device possession $\neq$ proven exclusive use

7. Professional Conclusion

The packet evidence establishes, to a high degree of confidence, that the harassing message received at lilytuckrige@yahoo.com on 22 July 2008 was submitted through willselfdestruct.com by a single, continuously identifiable device on the dorm network, and that the same device is strongly associated with the account jcoachj@gmail.com. The attribution chain is built from several independent layers of evidence that each address a different part of the “who did this” question, and each layer is anchored to a specific, reproducible packet or set of packets rather than to inference from the original complaint alone. First, the original e-mail's originating IP address was a shared student-accommodation address and was correctly treated in this investigation as a lead rather than an identification — a router or a NAT gateway serving many residents cannot itself be a suspect. The investigation therefore moved to packet-level analysis of the dorm uplink capture, where the second incident (the willselfdestruct.com submission) could be reconstructed end to end: DNS resolution of the site, a GET of the submission form, rendering of the form's supporting images, and then, 39.7 seconds later, an HTTP POST to /secure/submit whose form fields (to, subject, message) were recovered in full clear text from frame 83601. The recipient address in that POST, lilytuckrige@yahoo.com, is an exact match for the victim's reported address, and the message text (“Stop teaching. Start running.”) is consistent with the described harassment, which is why this specific POST — rather than any other HTTP traffic in the capture — is treated as the harassing submission. Second, that POST was tied to a specific network interface: Ethernet source address 00:17:f2:e2:c0:ce, an Apple-manufactured NIC. This is device-level, not person-level, attribution, but it is meaningfully stronger than the IP-only lead the case started with, because a MAC address is local-segment-specific and, in this capture, that MAC generated 77.6% of all traffic across the full 4-hour-22-minute window — the profile of one actively used personal device rather than a shared or transient one. Third, and most importantly for moving from a device to a person, the same MAC address repeatedly transmitted two independent, service-specific authenticated-session cookies — a Gmail chat-presence cookie and a separate Outlook/Live calendar cookie — both carrying the identifier jcoachj@gmail.com, across 149 matching frames spread through the capture. Two unrelated web services independently reflecting the same account on the same device is materially stronger corroboration than a single cookie sighting would be, because it is not merely one site's session state but a pattern repeated across the user's broader browsing on that machine. Fourth, that identifier was compared against the supplied Chemistry 109 roster. Under a standard first-initial-plus-surname e-mail convention, jcoachj corresponds uniquely to Johnny Coach, the only “Coach” on the roster and the only name that matches the recovered address under any reasonable construction. This step is circumstantial — it is a naming-pattern inference, not a verified account-registry lookup — and is weighted as such in §6.2. Taken together, the chain runs: complaint (shared IP, weak) → confirmed site and confirmed harassing POST content (strong, direct) → confirmed client MAC for that POST (strong, device-level) → strong cross-service identity correlation on that MAC (strong, but not absolute proof of exclusive use) → circumstantial roster name match (moderate). Because the strongest links in the chain (the POST content and the MAC/cookie correlation) are direct packet evidence and not inference, while the weakest link (the roster name match) is corroborating rather than load-bearing, the overall confidence in attributing the willselfdestruct.com submission to the device used by, and the account belonging to, Johnny Coach is HIGH for device/account attribution and

MODERATE-HIGH for the final human identification, given the open/shared nature of the dorm network and the absence of an independent, out-of-band confirmation (e.g. a device-possession or login-audit record) that no other person used this specific machine at 06:04 UTC on 22 July 2008.

Appendix A: Investigator Activity Log

Times below are read directly from the analysis workstation's on-screen clock (top-right of each session screenshot). They record when each examination step was performed during the review session and are not synchronised to, or comparable with, the 2008 UTC incident timeline in §5.1 — no date or timezone metadata was visible for the workstation clock, so these entries are presented as an internal, relative sequence only.

Time	Action	Ref.
14:42	Preserved the assigned evidence file and computed a cryptographic hash prior to any analysis: <code>openssl dgst -md5 nitroba.pcap</code> .	Fig. A-1
14:46	Opened the hashed working copy in Wireshark; performed an initial scan of the end of the capture to confirm scope, protocols and endpoint activity.	Fig. A-2
14:52	Applied filter <code>http.request.method=="GET" && http.host=="www.willselfdestruct.com"</code> ; located the client's GET /secure/submit request (frame 82936).	Fig. A-3
14:55	Reviewed the corresponding HTTP 200 OK response (frame 82998), noting the Set-Cookie: JSESSIONID value and Apache server banner.	Fig. A-4
15:01	Applied filter <code>(ip.src==192.168.15.4) && (ip.dst==69.25.94.22) && http.request.method=="POST"</code> ; located and decoded frame 83601, recovering the submitted form fields.	Fig. A-5
15:04	Expanded the Ethernet II layer of frame 83601 to record source/destination hardware addresses.	Fig. A-6
15:08	Applied filter <code>eth.addr==00:17:f2:e2:c0:ce</code> to assess the proportion of total capture traffic associated with the client MAC.	Fig. A-7
15:11	Applied filter <code>eth.src==00:17:f2:e2:c0:ce && http.cookie contains "@"</code> ; identified a Gmail chat-presence cookie carrying an e-mail-derived identifier.	Fig. A-8
15:18	Set Wireshark's time display format to UTC Date and Time of Day to prepare for timestamp normalisation.	Fig. A-9
15:23	Located the POST's corresponding server confirmation response containing the phrase "secure anonymous E-mail".	Fig. A-10
15:48	(TShark, terminal) Re-confirmed acquisition provenance (digitalcorpora.org 2008-nitroba corpus) and re-verified the MD5 hash of the working file.	Fig. A-11
15:50	<code>tshark -r nitroba.pcap head/tail</code> — confirmed first and last frame numbers and summary content.	Fig. A-12
15:51	<code>tshark -qz io,stat</code> — confirmed total frame count (94,410) and capture duration (15,760.0 s).	Fig. A-13
15:52	<code>tshark -T fields -e frame.time -e frame head</code> — recorded the absolute local timestamp of frame 1 (capture start).	Fig. A-14
15:54	<code>tshark -Y (GET filter)</code> — produced the full chronological sequence of requests to <code>www.willselfdestruct.com</code> .	Fig. A-15
15:55	Re-ran the GET-sequence query with -T fields (frame.time, frame.time_delta_displayed, ip.src, ip.dst, http.request.uri) for an exact, exportable timeline.	Fig. A-16
15:56	<code>tshark -Y "data-text-lines matches..."</code> — isolated the two HTTP responses containing the confirmation-page text.	Fig. A-17
16:03	<code>tshark -Y (POST filter) and frame 83601 -V grep</code> — cross-verified the POST content and confirmed the "to" field against the victim's reported address.	Fig. A-18
16:05	<code>tshark -T fields -e frame.time -e text</code> — extracted the POST's exact timestamp together with the full decoded form-field text in one query.	Fig. A-19

Time	Action	Ref.
16:06	Re-ran frame 83601 -V grep "form item" for a clean, isolated listing of all nine submitted form fields.	Fig. A-20
16:09	Confirmed the urlencoded-form.key / urlencoded-form.value TShark field definitions used for the structured extraction.	Fig. A-21
16:13	Confirmed eth.src for frame 83601 and quantified total frames/bytes associated with that MAC via io,stat.	Fig. A-22
16:15	Counted frames from the client MAC containing an "@"-bearing cookie value (149 frames).	Fig. A-23
16:16	Ran a regex extraction (-V grep -P) across all matching frames to enumerate every recovered e-mail-pattern cookie value from the client MAC.	Fig. A-24
16:17	Re-confirmed the exact POST frame timestamp for cross-reference in the integrated timeline.	Fig. A-25
—	Reviewed the supplied Chemistry 109 class-roster slide for comparison against the recovered identity artefact.	Fig. A-26

Appendix B: Evidence Register

ID	Description	Type	Source / provenance	Hash
EV-01	nitroba.pcap dorm Ethernet uplink capture	Primary digital evidence	digitalcorpora.org, 2008-nitroba public training corpus (downloaded via wget)	MD5: 9981827f11968773ff815e39f5458ec8
EV-02	Chem 109 Class roster	Reference/comparison evidence	Supplied as part of the case package (slide)	Not separately hashed supplied as a static reference slide artefact
EV-03	Case_Scenario.docx supplied case package (brief, module text, 26 analyst screenshots)	Case documentation / secondary evidentiary record	Supplied by module assessor	SHA-256: 1578cd468ff52fdc724b08490c640ee3efefd99e3e50651 MD5: 88efea31e0b450c97d93d39e5a6edab8

EV-01 is the sole item of primary technical evidence in this examination; all findings in §4 trace to it. EV-02 and EV-03 are reference/documentary items used for comparison and provenance only.

Appendix C: Hash Values

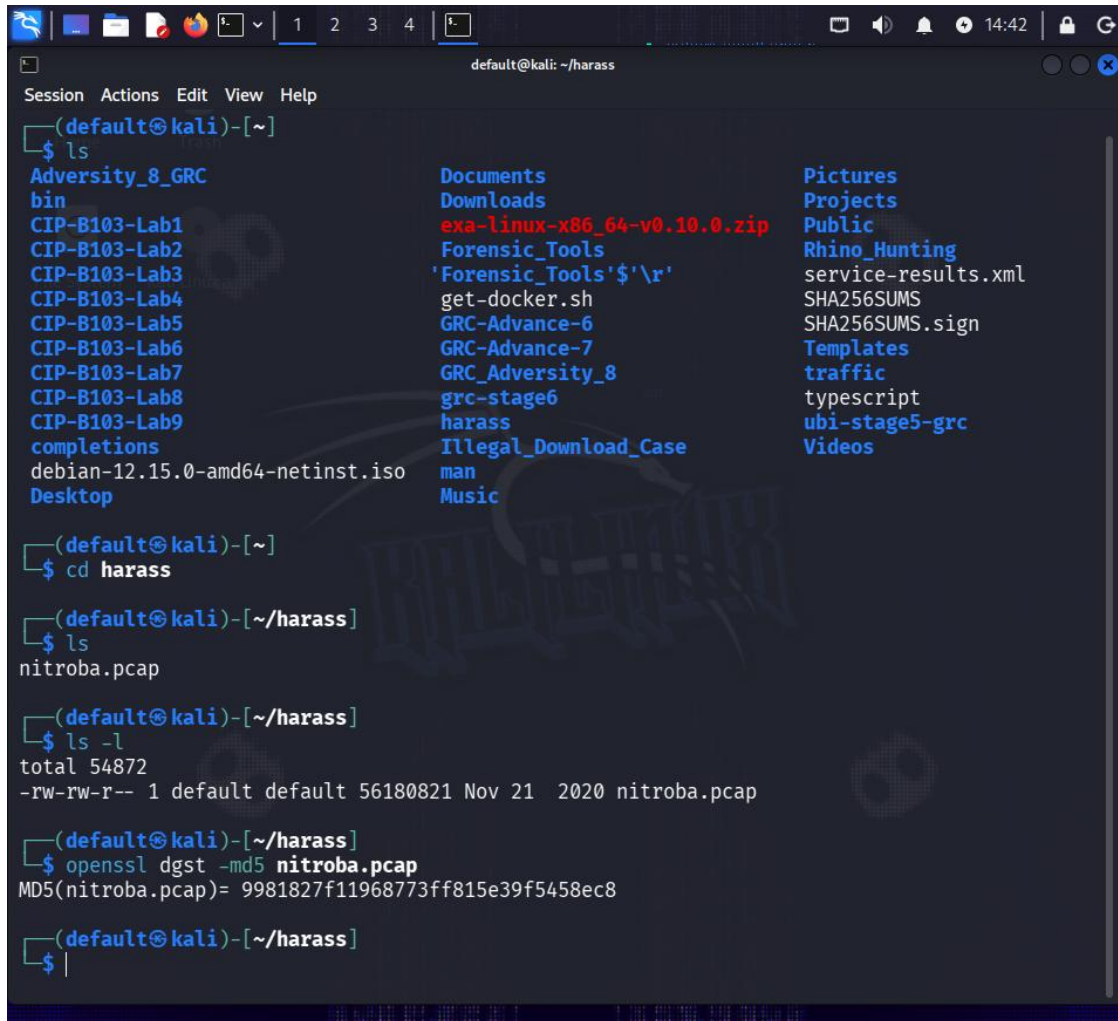
Exhibit	Algorithm	Value	Computed by
EV-01 nitroba.pcap	MD5	9981827f11968773ff815e39f5458ec8	openssl dgst -md5 nitroba.pcap (recorded at acquisition, re-verified mid-session — Appendix A, 15:48)
EV-03 Case_Scenario.docx	SHA-256	1578cd468ff52fdc724b08490c640ee3efefd99e3e5065bfcc1afaf7705e14fb	sha256sum (computed for this report, on the case package as supplied)
EV-03 Case_Scenario.docx	MD5	88efea31e0b450c97d93d39e5a6edab8	md5sum (computed for this report, on the case package as supplied)

Appendix D: Commands and Queries

Every command below was run against the hash-verified EV-01 working copy and is reproducible by re-opening the file with the same command.

Purpose	Command / filter
Hash the evidence file	<code>openssl dgst -md5 nitroba.pcap</code>
Capture summary (frame count, duration)	<code>tshark -r nitroba.pcap -qz io,stat,0,"COUNT(frame) frame"</code>
First/last frame timestamps	<code>tshark -r nitroba.pcap -T fields -e frame.time -e frame head -n 3 / tail -n 2</code>
Site access sequence	<code>tshark -r nitroba.pcap -Y 'http.request.method=="GET" && http.host=="www.willselfdestruct.com"'</code>
Site access with absolute/relative time	<code>tshark -r nitroba.pcap -Y '...' -T fields -e frame.time -e frame.time_delta_displayed -e ip.src -e ip.dst -e http.request.uri</code>
Isolate the harassment POST	<code>tshark -r nitroba.pcap -Y '(ip.src==192.168.15.4) && (ip.dst==69.25.94.22) && http.request.method=="POST"'</code>
Decode POST form fields	<code>tshark -r nitroba.pcap -Y "frame.number==83601" -V grep -i "form item"</code>
POST fields as a flat record	<code>tshark -r nitroba.pcap -Y "frame.number==83601" -T fields -e urlencoded-form.key -e urlencoded-form.value</code>
POST timestamp + full decoded text in one query	<code>tshark -r nitroba.pcap -Y "(ip.src==192.168.15.4)&&(ip.dst==69.25.94.22)&&http.request.method=="POST\" -T fields -e frame.time -e text</code>
Client MAC for the POST frame	<code>tshark -r nitroba.pcap -Y "frame.number==83601" -T fields -e eth.src</code>
Volume of traffic from that MAC	<code>tshark -r nitroba.pcap -qz io,stat,0,"COUNT(frame) frame","eth.addr==00:17:f2:e2:c0:ce"</code>
Identity-bearing cookies from that MAC	<code>tshark -r nitroba.pcap -Y 'eth.src==00:17:f2:e2:c0:ce && http.cookie contains "@"'</code>
Enumerate every recovered e-mail-pattern cookie	<code>tshark -r nitroba.pcap -Y "eth.src==00:17:f2:e2:c0:ce" -V grep -P "[\\w.%+-]+@[\\w.-]+\\.\\.[a-zA-Z]{2,4}"</code>
Confirm the POST server response text	<code>tshark -r nitroba.pcap -Y 'data-text-lines matches "secure anonymous E-mail"'</code>
Re-verify evidence provenance / hash mid-session	<code>wget -q https://downloads.digitalcorpora.org/corpora/network-package-dumps/2008-nitroba/nitroba.pcap; openssl dgst -md5 nitroba.pcap</code>
Report-preparation: hash the supplied case package	<code>sha256sum Case_Scenario.docx; md5sum Case_Scenario.docx</code>

Appendix E: Screenshots



```

default@kali: ~/harass
Session Actions Edit View Help
(default@kali)-[~]
$ ls
Adversity_8_GRC      Documents            Pictures
bin                  Downloads           Projects
CIP-B103-Lab1        exa-linux-x86_64-v0.10.0.zip  Public
CIP-B103-Lab2        Forensic_Tools      Rhino_Hunting
CIP-B103-Lab3        'Forensic_Tools'\r'  service-results.xml
CIP-B103-Lab4        get-docker.sh       SHA256SUMS
CIP-B103-Lab5        GRC-Advance-6       SHA256SUMS.sign
CIP-B103-Lab6        GRC-Advance-7       Templates
CIP-B103-Lab7        GRC_Adversity_8     traffic
CIP-B103-Lab8        grc-stage6          typescript
CIP-B103-Lab9        harass              ubi-stage5-grc
completions          Illegal_Download_Case  Videos
debian-12.15.0-amd64-netinst.iso
Desktop              man
                    Music

(default@kali)-[~]
$ cd harass

(default@kali)-[~/harass]
$ ls
nitroba.pcap

(default@kali)-[~/harass]
$ ls -l
total 54872
-rw-rw-r-- 1 default default 56180821 Nov 21 2020 nitroba.pcap

(default@kali)-[~/harass]
$ openssl dgst -md5 nitroba.pcap
MD5(nitroba.pcap)= 9981827f11968773ff815e39f5458ec8

(default@kali)-[~/harass]
$
  
```

Fig. A-1 (14:42) — Evidence preserved and hashed: `openssl dgst -md5 nitroba.pcap`.

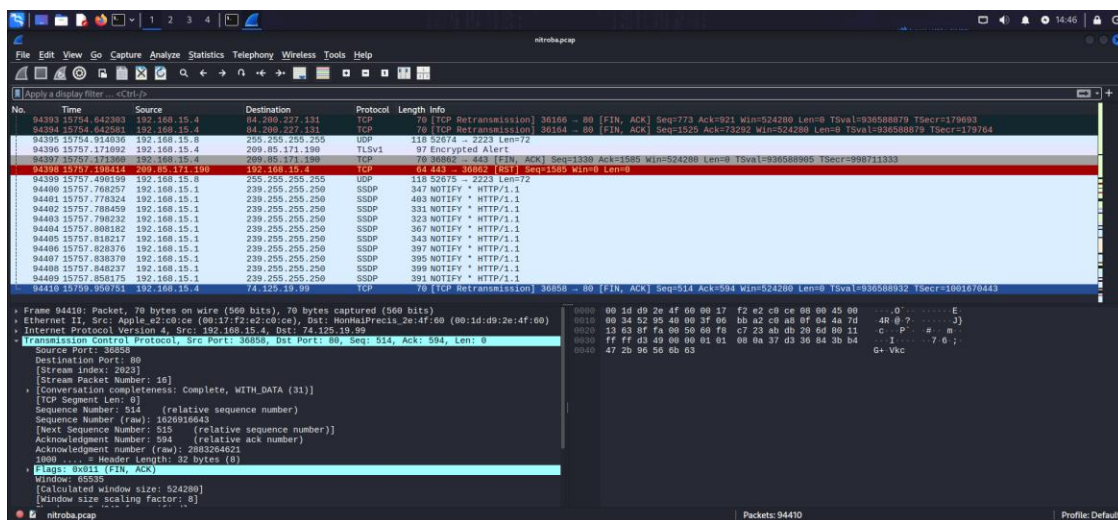


Fig. A-2 (14:46) — Initial Wireshark scan of the end of the capture (scope/endpoint check).

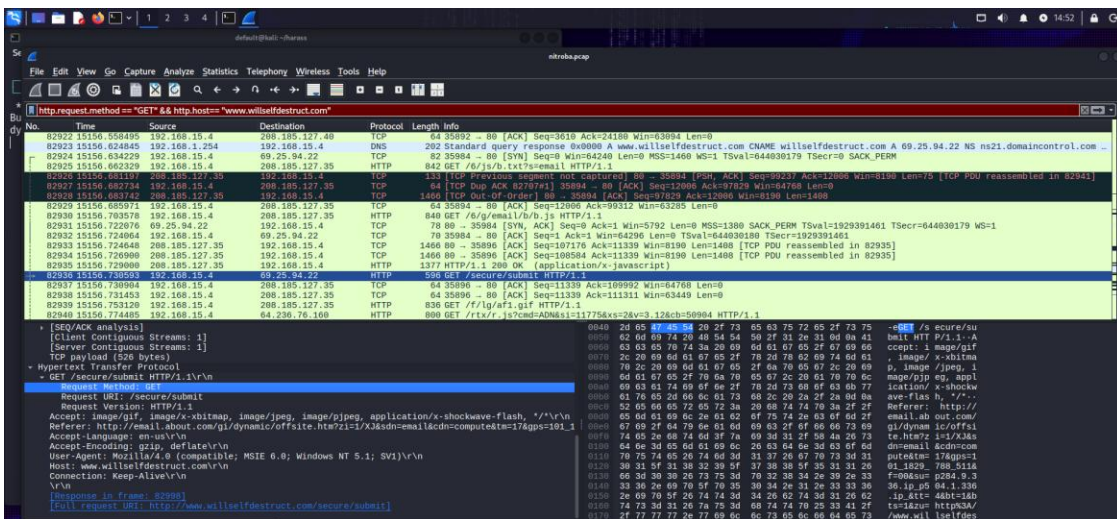


Fig. A-3 (14:52) — GET filter isolates frame 82936, GET /secure/submit.

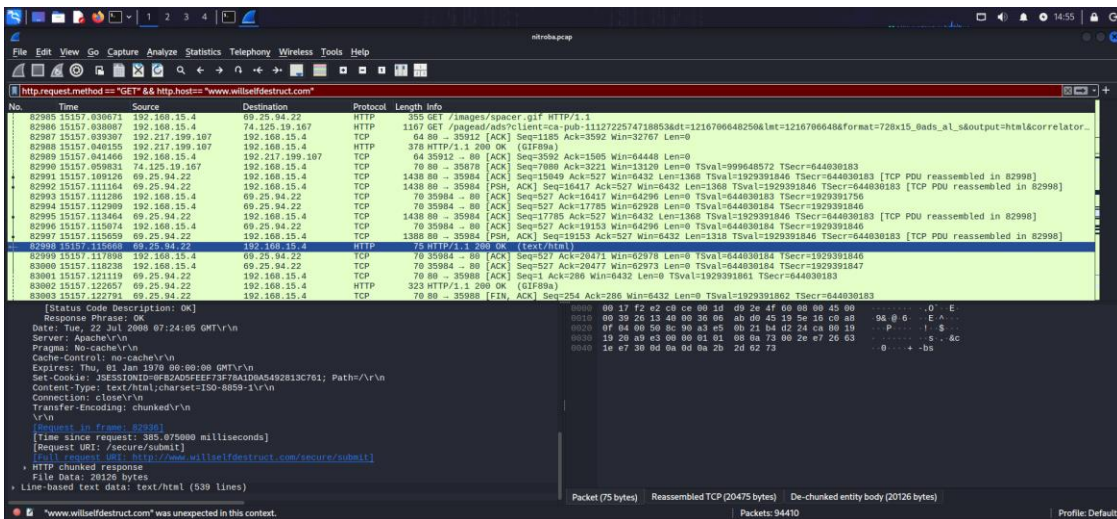


Fig. A-4 (14:55) — Server response to the form GET: Set-Cookie JSESSIONID, Apache banner.

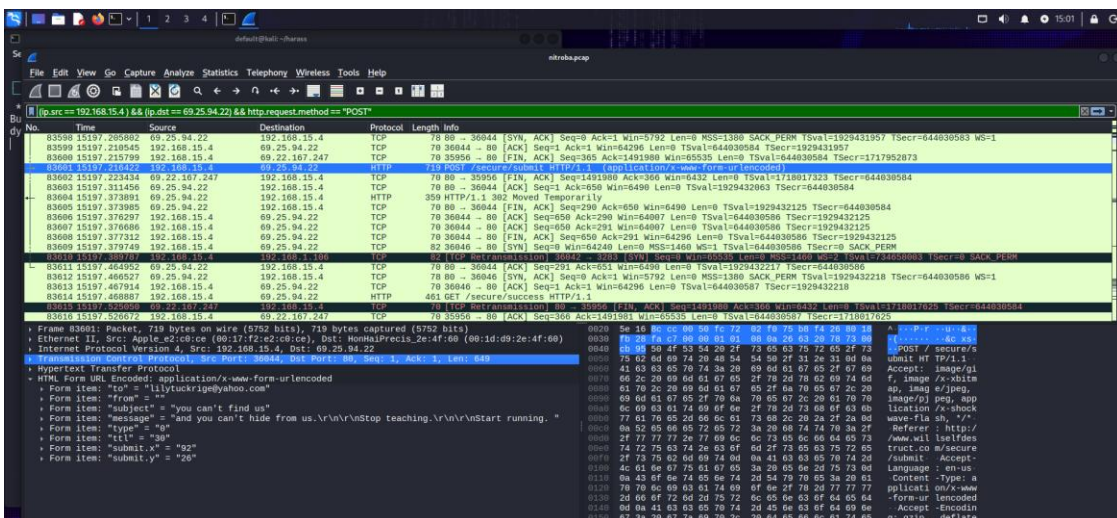


Fig. A-5 (15:01) — Frame 83601 decoded: the harassing POST and its form fields.

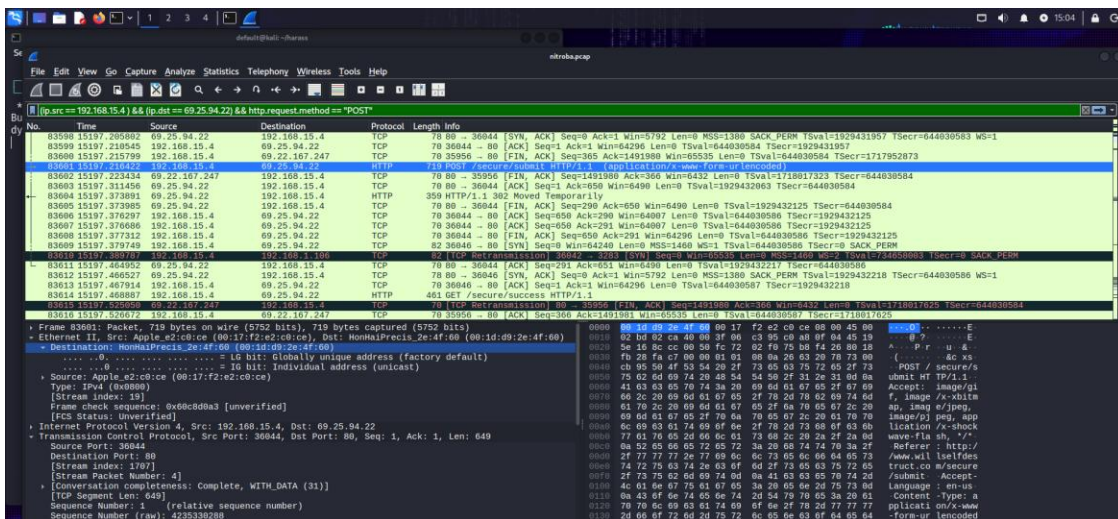


Fig. A-6 (15:04) — Ethernet II layer of frame 83601: source/destination MAC addresses.

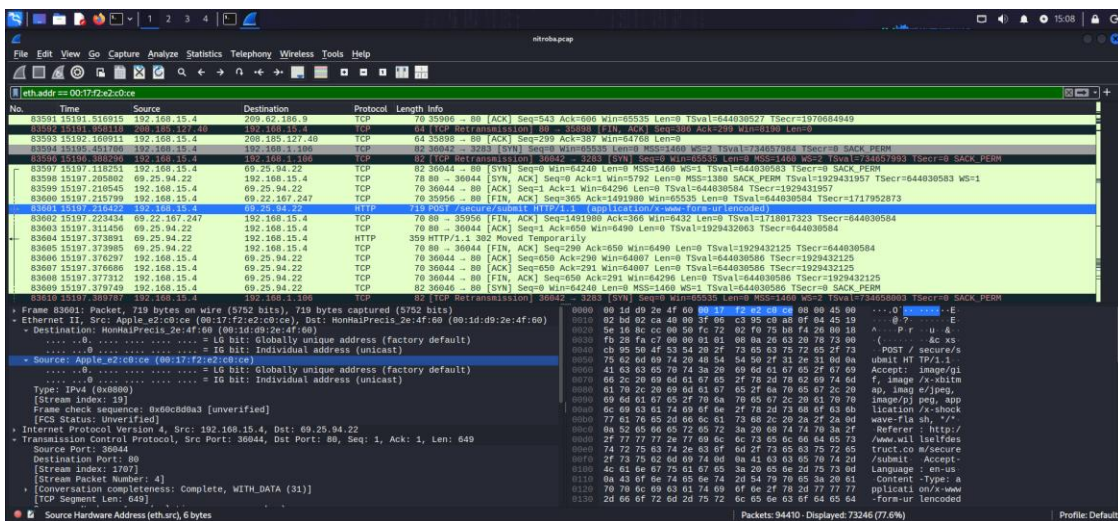


Fig. A-7 (15:08) — eth.addr filter quantifies traffic volume from the client MAC (73,246 frames, 77.6%).

eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"

No.	Time	Source	Destination	Protocol	Length	Info
77528	14977.375246	192.168.15.4	74.125.19.104	HTTP	1463	GET /calendar/e0e03af5825c
77547	14977.571296	192.168.15.4	74.125.19.104	HTTP	81	GET /calendar/e0e03af5825c
77710	14978.265471	192.168.15.4	74.125.19.104	HTTP	151	POST /calendar/calddetails
77711	14978.265791	192.168.15.4	74.125.19.104	HTTP	336	POST /calendar/load HTTP/1
77727	14983.286302	192.168.15.4	74.125.19.104	HTTP	84	GET /calendar/e0e03af5825c
78967	14995.018158	192.168.15.4	74.125.19.17	HTTP	1345	GET /mail/channel/test?at=
78968	14995.019816	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78975	14995.242111	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78977	14995.282628	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78984	14995.371518	192.168.15.4	209.85.201.189	HTTP	1359	GET /mail/images/cleardot.
78990	14995.451768	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78996	14995.493422	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
79007	14995.666665	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
79012	14995.705585	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
79014	14995.728533	192.168.15.4	209.85.201.189	HTTP	92	GET /mail/channel/test?at=
79019	14995.880621	192.168.15.4	74.125.19.17	HTTP	1307	GET /mail/?ui=1&view=page8
79023	14995.928149	192.168.15.4	74.125.19.17	HTTP	1301	GET /mail/help/images/logic
79032	14995.984476	192.168.15.4	74.125.19.17	HTTP	1303	GET /mail/images/star_on_s
79036	14996.088914	192.168.15.4	74.125.19.17	HTTP	1306	GET /mail/images/chat_bubb

Accept-Encoding: gzip, deflate\r\n
 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
 Host: mail.google.com\r\n
 Connection: Keep-Alive\r\n

Cookie pair: GX=DQAAAG8AAAAm2oW8LqM60qoQ5w2jVJ-ZHIFuyAQ3GUKvcv4N9vQ6lWuLpVMCmw1Jhm1m9_P3qZbyTwkIWd...
 Cookie pair: S=gmail=L5bh7hHJ9B97n6StWA4FvA:gmil_yj=-OoenmU7qTeuQ1dsN3B1kg:gmproxy=6uatNcZzM8:...
 Cookie pair: GMAIL_AT=xn3j32oktf2a0q6oa3k9sfr6d09yzf
 Cookie pair: GMAIL_SU=1
 Cookie pair: gmailchat=jcoachj@gmail.com/475090
 Cookie pair: PREF=ID=8fc081df5e738a3c:TM=1210743469:LM=1210743469:S=PiBsyJkS6cu-UUEXV
 Cookie pair: NID=13=tJ7LEc6z2iH48P_IPyV0gGh14alcZoJcJaf7l-9JQ2AeoD8oWG9NjT0p7T5tuskkNgEKMRA9P...
 Cookie pair: __utmz=173272373.00000983192309928271:2:
 Cookie pair: __utmz=173272373.00000983192309928271:1216706401:2592000
 Cookie pair: SID=DQAAAGwAAACH8Y_j5izp1fDbDjZwRDFDgtU3aaeZKwGZ7DwUjYpLoqH7F1_E-X5taC4l0uvzXtrVeE6...
 Cookie pair: TZ=-60
 Cookie pair: GMAIL_HELP=hosted:0

Response in frame: 78973
 Full request URI: http://mail.google.com/mail/?ui=1&view=page&name=gp&ver=sh3fb53pgpk&auto=1]

A name/value HTTP cookie pair (http.cookie.pair) 34 bytes

Fig. A-8 (15:11) — Cookie filter recovers gmailchat=jcoachj@gmail.com from the client MAC.

Filter: eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"

No.	Time	Source	Destination	Protocol	Length	Info
77528	14977.375246	192.168.15.4	74.125.19.104	HTTP	1463	GET /calendar/e0e03af5825c
77547	14977.571296	192.168.15.4	74.125.19.104	HTTP	81	GET /calendar/e0e03af5825c
77710	14978.265471	192.168.15.4	74.125.19.104	HTTP	151	POST /calendar/calddetails
77711	14978.265791	192.168.15.4	74.125.19.104	HTTP	336	POST /calendar/load HTTP/1
77727	14983.286302	192.168.15.4	74.125.19.104	HTTP	84	GET /calendar/e0e03af5825c
78967	14995.018158	192.168.15.4	74.125.19.17	HTTP	1345	GET /mail/channel/test?at=
78968	14995.019816	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78975	14995.242111	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78977	14995.282628	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78984	14995.371518	192.168.15.4	209.85.201.189	HTTP	1359	GET /mail/images/cleardot.
78990	14995.451768	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
78996	14995.493422	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
79007	14995.666665	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
79012	14995.705585	192.168.15.4	74.125.19.17	HTTP	1391	GET /mail/?ui=1&view=page8
79014	14995.728533	192.168.15.4	209.85.201.189	HTTP	92	GET /mail/channel/test?at=
79019	14995.880621	192.168.15.4	74.125.19.17	HTTP	1307	GET /mail/?ui=1&view=page8
79023	14995.928149	192.168.15.4	74.125.19.17	HTTP	1301	GET /mail/help/images/logic
79032	14995.984476	192.168.15.4	74.125.19.17	HTTP	1303	GET /mail/images/star_on_s
79036	14996.088914	192.168.15.4	74.125.19.17	HTTP	1306	GET /mail/images/chat_bubb

Accept-Encoding: gzip, deflate\r\n
 User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
 Host: mail.google.com\r\n
 Connection: Keep-Alive\r\n

Cookie pair: GX=DQAAAG8AAAAm2oW8LqM60qoQ5w2jVJ-ZHIFuyAQ3GUKvcv4N9vQ6lWuLpVMCmw1Jhm1m9_P3qZbyTwkIWd...
 Cookie pair: S=gmail=L5bh7hHJ9B97n6StWA4FvA:gmil_yj=-OoenmU7qTeuQ1dsN3B1kg:gmproxy=6uatNcZzM8:...
 Cookie pair: GMAIL_AT=xn3j32oktf2a0q6oa3k9sfr6d09yzf
 Cookie pair: GMAIL_SU=1
 Cookie pair: gmailchat=jcoachj@gmail.com/475090
 Cookie pair: PREF=ID=8fc081df5e738a3c:TM=1210743469:LM=1210743469:S=PiBsyJkS6cu-UUEXV
 Cookie pair: NID=13=tJ7LEc6z2iH48P_IPyV0gGh14alcZoJcJaf7l-9JQ2AeoD8oWG9NjT0p7T5tuskkNgEKMRA9P...
 Cookie pair: __utmz=173272373.00000983192309928271:2:
 Cookie pair: __utmz=173272373.00000983192309928271:1216706401:2592000
 Cookie pair: SID=DQAAAGwAAACH8Y_j5izp1fDbDjZwRDFDgtU3aaeZKwGZ7DwUjYpLoqH7F1_E-X5taC4l0uvzXtrVeE6...
 Cookie pair: TZ=-60
 Cookie pair: GMAIL_HELP=hosted:0

Response in frame: 78973
 Full request URI: http://mail.google.com/mail/?ui=1&view=page&name=gp&ver=sh3fb53pgpk&auto=1]

A name/value HTTP cookie pair (http.cookie.pair) 34 bytes

Fig. A-9 (15:18) — Wireshark time-display format set to UTC for timestamp normalisation.

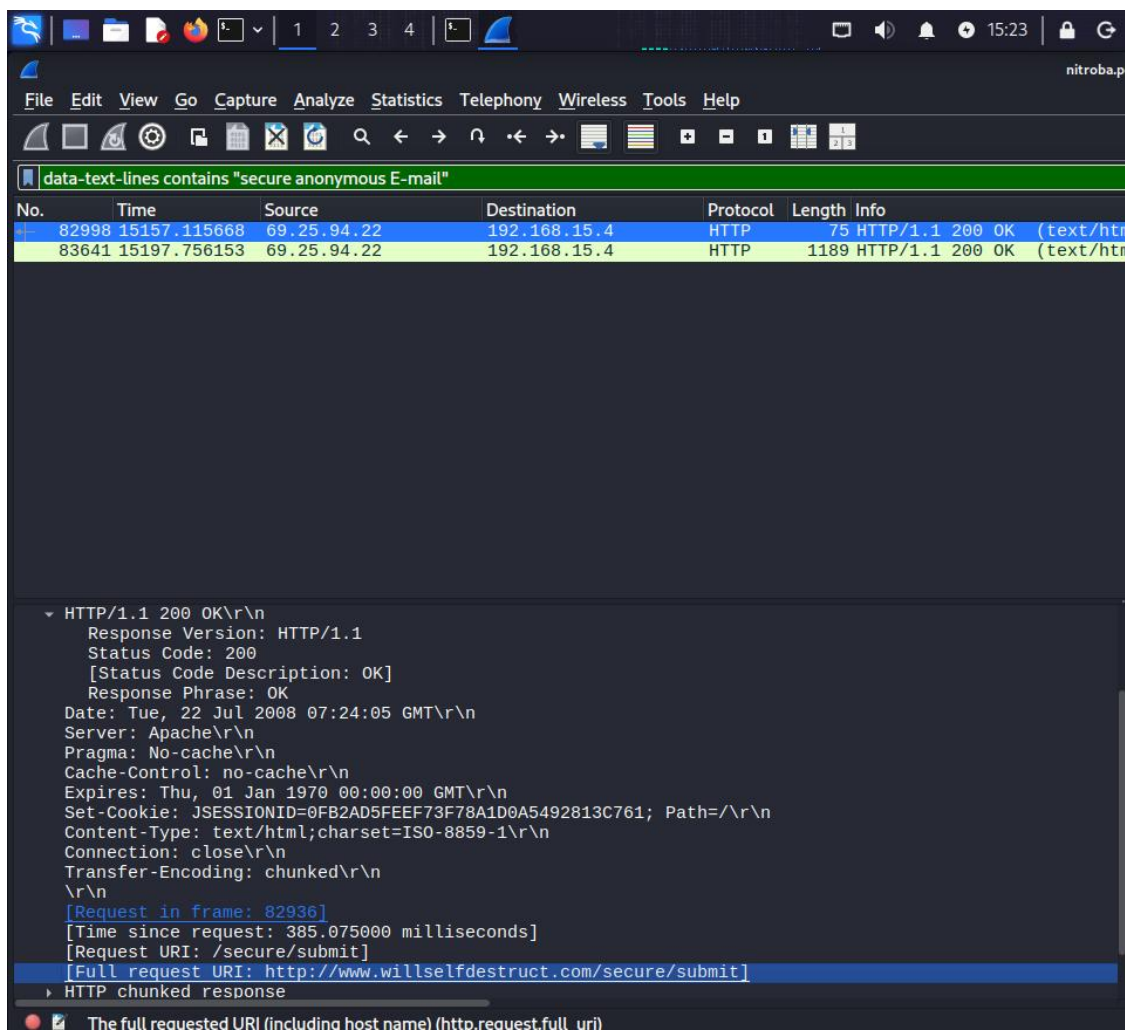
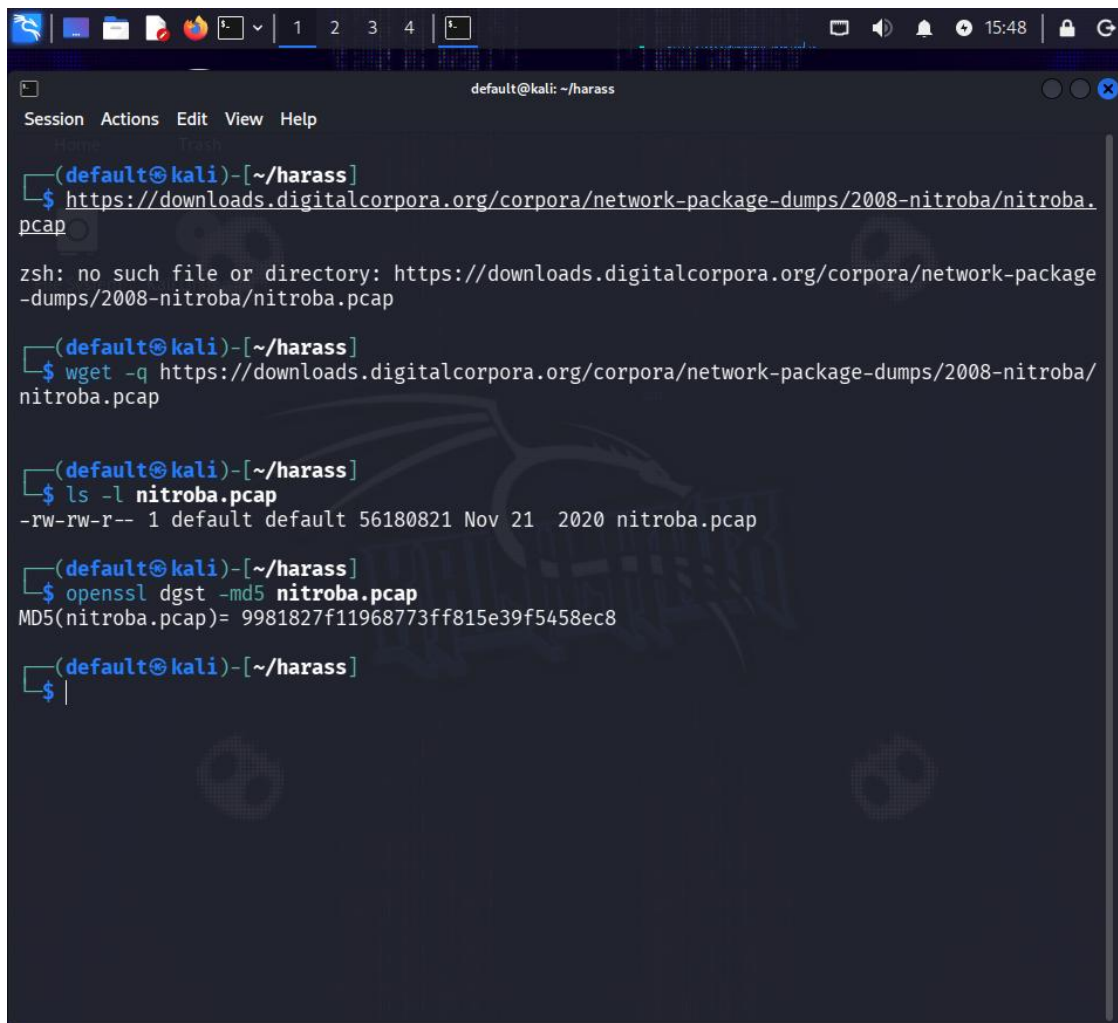


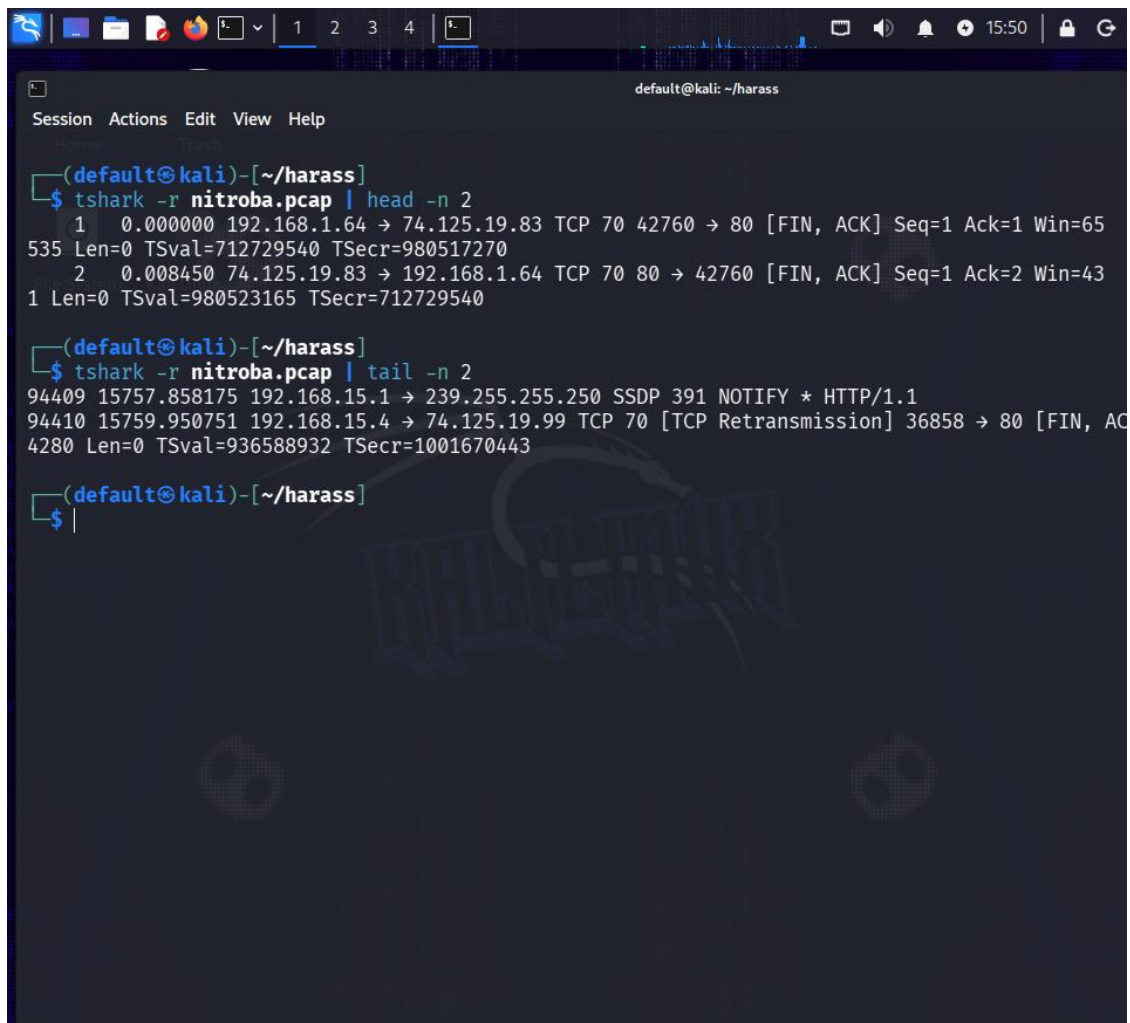
Fig. A-10 (15:23) — Server confirmation response containing “secure anonymous E-mail” text.



The image shows a terminal window on a Kali Linux system. The user is in the directory ~/harass. They attempt to download a file from a URL using a command that is partially obscured. The download fails with a 'zsh: no such file or directory' error. The user then uses 'wget' to successfully download the file 'nitroba.pcap'. They verify the file's presence with 'ls -l' and calculate its MD5 hash using 'openssl dgst -md5'. The terminal output shows the file was downloaded and its MD5 hash is 9981827f11968773ff815e39f5458ec8.

```
default@kali: ~/harass
Session Actions Edit View Help
Home Trash
(default@kali)~[~/harass]
$ https://downloads.digitalcorpora.org/corpora/network-package-dumps/2008-nitroba/nitroba.pcap
zsh: no such file or directory: https://downloads.digitalcorpora.org/corpora/network-package-dumps/2008-nitroba/nitroba.pcap
(default@kali)~[~/harass]
$ wget -q https://downloads.digitalcorpora.org/corpora/network-package-dumps/2008-nitroba/nitroba.pcap
(default@kali)~[~/harass]
$ ls -l nitroba.pcap
-rw-rw-r-- 1 default default 56180821 Nov 21 2020 nitroba.pcap
(default@kali)~[~/harass]
$ openssl dgst -md5 nitroba.pcap
MD5(nitroba.pcap)= 9981827f11968773ff815e39f5458ec8
(default@kali)~[~/harass]
$
```

Fig. A-11 (15:48) — Terminal: evidence re-acquired from digitalcorpora.org; MD5 re-verified.



The screenshot shows a Kali Linux terminal window with the following content:

```
default@kali: ~/harass
Session Actions Edit View Help
Home Trash
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap | head -n 2
1 0.000000 192.168.1.64 → 74.125.19.83 TCP 70 42760 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65
535 Len=0 TSval=712729540 TSecr=980517270
2 0.008450 74.125.19.83 → 192.168.1.64 TCP 70 80 → 42760 [FIN, ACK] Seq=1 Ack=2 Win=43
1 Len=0 TSval=980523165 TSecr=712729540

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap | tail -n 2
94409 15757.858175 192.168.15.1 → 239.255.255.250 SSDP 391 NOTIFY * HTTP/1.1
94410 15759.950751 192.168.15.4 → 74.125.19.99 TCP 70 [TCP Retransmission] 36858 → 80 [FIN, AC
4280 Len=0 TSval=936588932 TSecr=1001670443

(default@kali)-[~/harass]
$
```

Fig. A-12 (15:50) — tshark head/tail confirms first and last frames of the capture.


```

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap | head -n 2
 1   0.000000 192.168.1.64 → 74.125.19.83 TCP 70 42760 → 80 [FIN, ACK] Seq=1 Ack=1 Win=65
535 Len=0 TSval=712729540 TSecr=980517270
 2   0.008450 74.125.19.83 → 192.168.1.64 TCP 70 80 → 42760 [FIN, ACK] Seq=1 Ack=2 Win=43
1 Len=0 TSval=980523165 TSecr=712729540

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap | tail -n 2
94409 15757.858175 192.168.15.1 → 239.255.255.250 SSDP 391 NOTIFY * HTTP/1.1
94410 15759.950751 192.168.15.4 → 74.125.19.99 TCP 70 [TCP Retransmission] 36858 → 80 [FIN, ACK] Seq=514 Ack=594 Win=52
4280 Len=0 TSval=936588932 TSecr=1001670443

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -qz io,stat,0,"COUNT(frame) frame"

IO Statistics
Duration: 15760.0 secs
Interval: 15760.0 secs
Col 1: COUNT(frame) frame

Interval | 1 | COUNT |
0.0 < 15760.0 | 94410 |

(default@kali)-[~/harass]
$

```

Fig. A-13 (15:51) — tshark io,stat confirms total frame count and capture duration.

```

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap | tail -n 2
94409 15757.858175 192.168.15.1 → 239.255.255.250 SSDP 391 NOTIFY * HTTP/1.1
94410 15759.950751 192.168.15.4 → 74.125.19.99 TCP 70 [TCP Retransmission] 36858 → 80 [FIN, ACK] Seq=514 Ack=594 Win=52
4280 Len=0 TSval=936588932 TSecr=1001670443

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -qz io,stat,0,"COUNT(frame) frame"

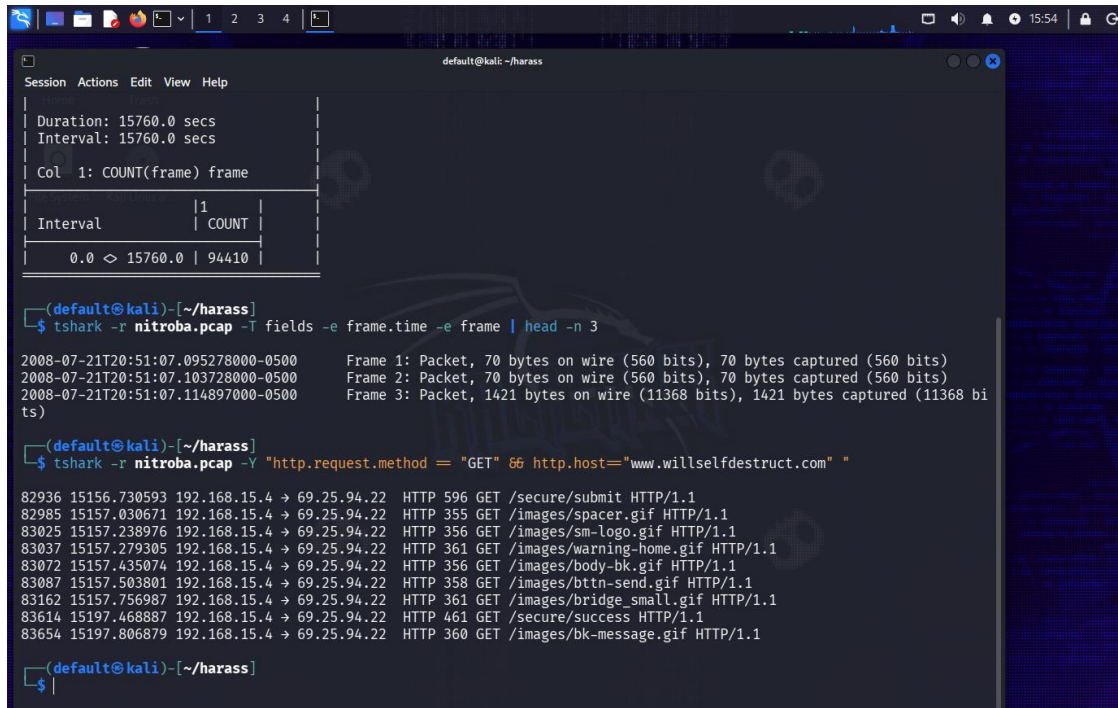
IO Statistics
Duration: 15760.0 secs
Interval: 15760.0 secs
Col 1: COUNT(frame) frame

Interval | 1 | COUNT |
0.0 < 15760.0 | 94410 |

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -T fields -e frame.time -e frame | head -n 3
2008-07-21T20:51:07.095278000-0500   Frame 1: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits)
2008-07-21T20:51:07.103728000-0500   Frame 2: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits)
2008-07-21T20:51:07.114897000-0500   Frame 3: Packet, 1421 bytes on wire (11368 bits), 1421 bytes captured (11368 bi
ts)

(default@kali)-[~/harass]
$

```

Fig. A-14 (15:52) — *tshark* fields extraction: absolute local timestamp of frame 1.


```

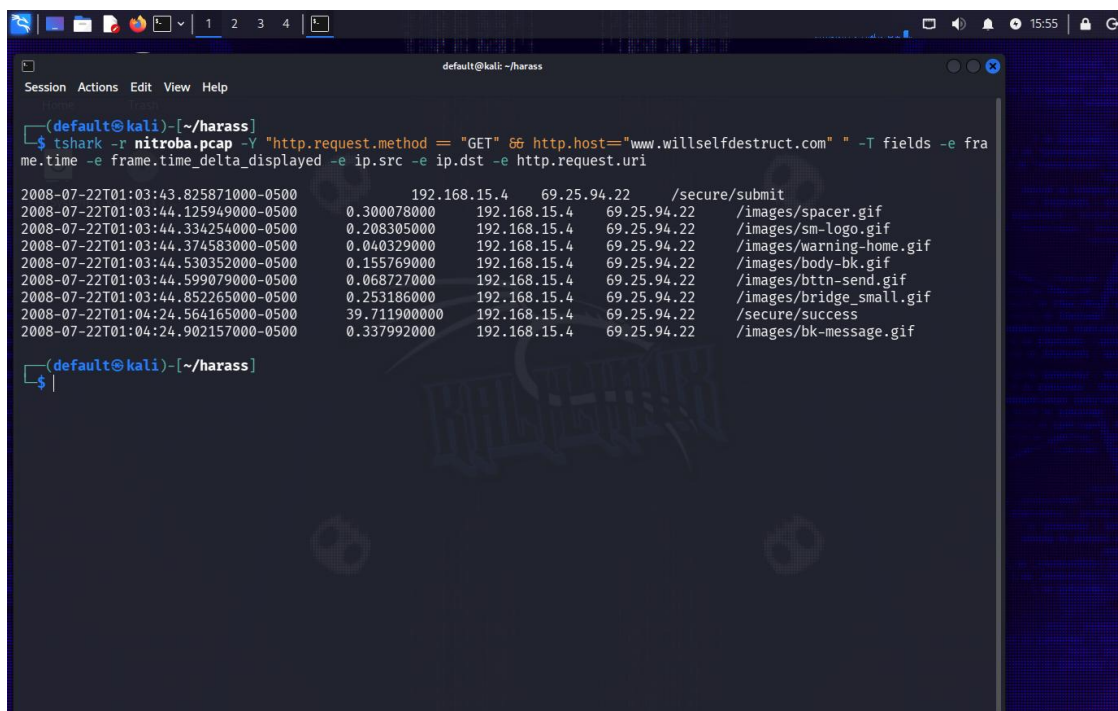
Session Actions Edit View Help
|-----|
| Duration: 15760.0 secs |
| Interval: 15760.0 secs |
| Col 1: COUNT(frame) frame |
|-----|
| Interval | 1 |
|-----|
| 0.0 <- 15760.0 | 94410 |
|-----|

(default@kali)~/harass
$ tshark -r nitroba.pcap -T fields -e frame.time -e frame | head -n 3
2008-07-21T20:51:07.095278000-0500      Frame 1: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits)
2008-07-21T20:51:07.103728000-0500      Frame 2: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits)
2008-07-21T20:51:07.114897000-0500      Frame 3: Packet, 1421 bytes on wire (11368 bits), 1421 bytes captured (11368 bi
ts)

(default@kali)~/harass
$ tshark -r nitroba.pcap -Y "http.request.method == \"GET\" && http.host==\"www.willselfdestruct.com\" "
82936 15156.730593 192.168.15.4 -> 69.25.94.22 HTTP 596 GET /secure/submit HTTP/1.1
82985 15157.030671 192.168.15.4 -> 69.25.94.22 HTTP 355 GET /images/spacer.gif HTTP/1.1
83025 15157.238976 192.168.15.4 -> 69.25.94.22 HTTP 356 GET /images/sm-logo.gif HTTP/1.1
83037 15157.279305 192.168.15.4 -> 69.25.94.22 HTTP 361 GET /images/warning-home.gif HTTP/1.1
83072 15157.435074 192.168.15.4 -> 69.25.94.22 HTTP 356 GET /images/body-bk.gif HTTP/1.1
83087 15157.503801 192.168.15.4 -> 69.25.94.22 HTTP 358 GET /images/btttn-send.gif HTTP/1.1
83162 15157.756987 192.168.15.4 -> 69.25.94.22 HTTP 361 GET /images/bridge_small.gif HTTP/1.1
83614 15197.468887 192.168.15.4 -> 69.25.94.22 HTTP 461 GET /secure/success HTTP/1.1
83654 15197.806879 192.168.15.4 -> 69.25.94.22 HTTP 360 GET /images/bk-message.gif HTTP/1.1

(default@kali)~/harass
$

```

Fig. A-15 (15:54) — *tshark* GET-filter results: full chronological site-access sequence.


```

Session Actions Edit View Help
|-----|
| Duration: 15760.0 secs |
| Interval: 15760.0 secs |
| Col 1: COUNT(frame) frame |
|-----|
| Interval | 1 |
|-----|
| 0.0 <- 15760.0 | 94410 |
|-----|

(default@kali)~/harass
$ tshark -r nitroba.pcap -Y "http.request.method == \"GET\" && http.host==\"www.willselfdestruct.com\" " -T fields -e fra
me.time -e frame.time_delta_displayed -e ip.src -e ip.dst -e http.request.uri
2008-07-22T01:03:43.825871000-0500      192.168.15.4      69.25.94.22      /secure/submit
2008-07-22T01:03:44.125949000-0500      0.300078000      192.168.15.4      69.25.94.22      /images/spacer.gif
2008-07-22T01:03:44.334254000-0500      0.208305000      192.168.15.4      69.25.94.22      /images/sm-logo.gif
2008-07-22T01:03:44.374583000-0500      0.040329000      192.168.15.4      69.25.94.22      /images/warning-home.gif
2008-07-22T01:03:44.530352000-0500      0.155769000      192.168.15.4      69.25.94.22      /images/body-bk.gif
2008-07-22T01:03:44.599079000-0500      0.068727000      192.168.15.4      69.25.94.22      /images/btttn-send.gif
2008-07-22T01:03:44.852265000-0500      0.253186000      192.168.15.4      69.25.94.22      /images/bridge_small.gif
2008-07-22T01:04:24.564165000-0500      39.711900000      192.168.15.4      69.25.94.22      /secure/success
2008-07-22T01:04:24.902157000-0500      0.337992000      192.168.15.4      69.25.94.22      /images/bk-message.gif

(default@kali)~/harass
$

```

Fig. A-16 (15:55) — *tshark* fields extraction: exact per-request timestamps for the GET sequence.

```

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "data-text-lines matches \\"secure anonymous E-mail\\"*" 2>/dev/null
82998 15157.115668 69.25.94.22 → 192.168.15.4 HTTP 75 HTTP/1.1 200 OK (text/html)
83641 15197.756153 69.25.94.22 → 192.168.15.4 HTTP 1189 HTTP/1.1 200 OK (text/html)
(default@kali)-[~/harass]
$

```

Fig. A-17 (15:56) — tshark filter isolates the two confirmation-text HTTP responses.

```

(default@kali)-[~/harass]
$ (ip.src == 192.168.15.4 ) && (ip.dst == 69.25.94.22) && http.request.method == "POST"
ip.src: command not found

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "(ip.src == 192.168.15.4 ) && (ip.dst == 69.25.94.22) && http.request.method == \"POST\" "
83601 15197.216422 192.168.15.4 → 69.25.94.22 HTTP 719 POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)

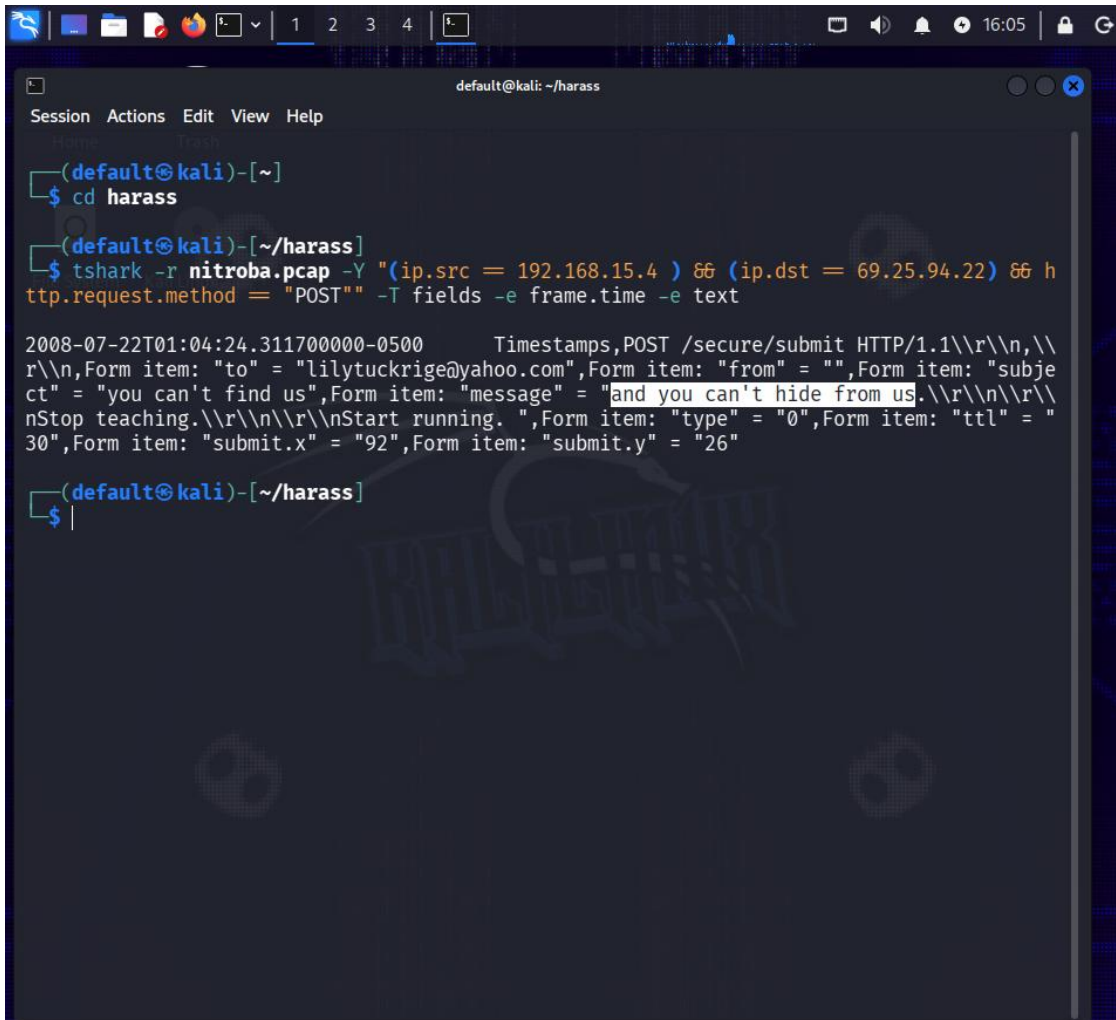
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "frame.number == 83601 -V | grep \"lilytuckrige@yahoo.com\""
dquote> tshark -r nitroba.pcap -Y "(ip.src == 192.168.15.4 ) && (ip.dst == 69.25.94.22) && http.request.method == \"POST\" "

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "frame.number == 83601" -v | grep "lilytuckrige@yahoo.com"

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "frame.number == 83601" -v | grep "lilytuckrige@yahoo.com"
Form item: "to" = "lilytuckrige@yahoo.com"
Value: lilytuckrige@yahoo.com

(default@kali)-[~/harass]
$

```

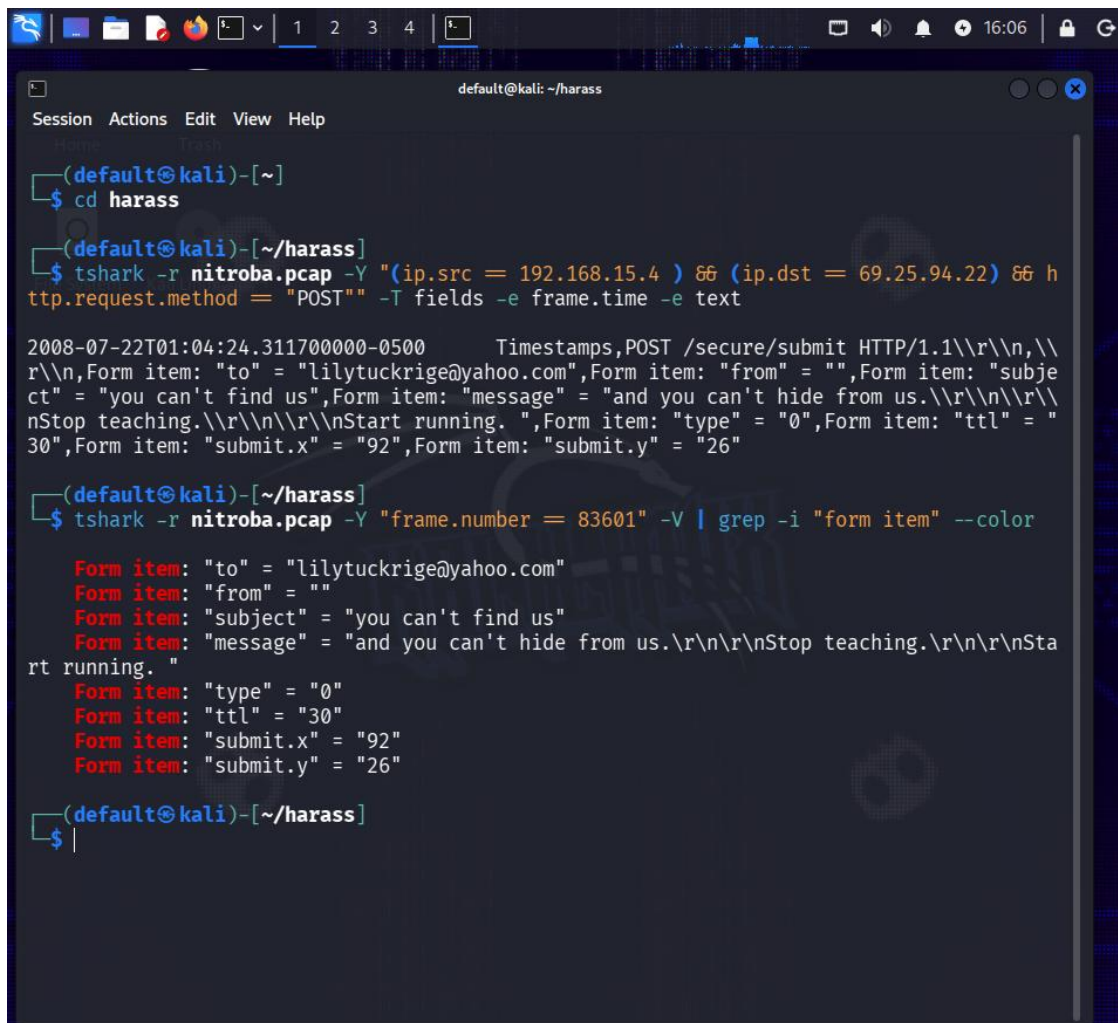

Fig. A-18 (16:03) — POST filter + frame 83601 grep cross-verifies the “to” field.

```
default@kali: ~/harass
Session Actions Edit View Help
home Trash
(default@kali)-[~]
$ cd harass
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "(ip.src == 192.168.15.4 ) && (ip.dst == 69.25.94.22) && http.request.method == \"POST\"" -T fields -e frame.time -e text

2008-07-22T01:04:24.311700000-0500      Timestamps,POST /secure/submit HTTP/1.1\r\n,\r\n,\r\n,Form item: "to" = "lilytuckrige@yahoo.com",Form item: "from" = "",Form item: "subject" = "you can't find us",Form item: "message" = "and you can't hide from us.\r\n\r\n\r\nStop teaching.\r\n\r\n\r\nStart running. ",Form item: "type" = "0",Form item: "ttl" = "30",Form item: "submit.x" = "92",Form item: "submit.y" = "26"

(default@kali)-[~/harass]
$
```

Fig. A-19 (16:05) — tshark fields extraction: exact POST timestamp with full decoded form text.



```

default@kali: ~/harass
Session Actions Edit View Help
(default@kali)-[~]
$ cd harass
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "(ip.src == 192.168.15.4 ) && (ip.dst == 69.25.94.22) && http.request.method == \"POST\" -T fields -e frame.time -e text

2008-07-22T01:04:24.311700000-0500      Timestamps,POST /secure/submit HTTP/1.1\\r\\n,\\r\\n,\\n,Form item: \"to\" = \"lilytuckrige@yahoo.com\",Form item: \"from\" = \"\",Form item: \"subject\" = \"you can't find us\",Form item: \"message\" = \"and you can't hide from us.\\r\\n\\r\\n\\r\\nStop teaching.\\r\\n\\n\\r\\nStart running. \",Form item: \"type\" = \"0\",Form item: \"ttl\" = \"30\",Form item: \"submit.x\" = \"92\",Form item: \"submit.y\" = \"26\"

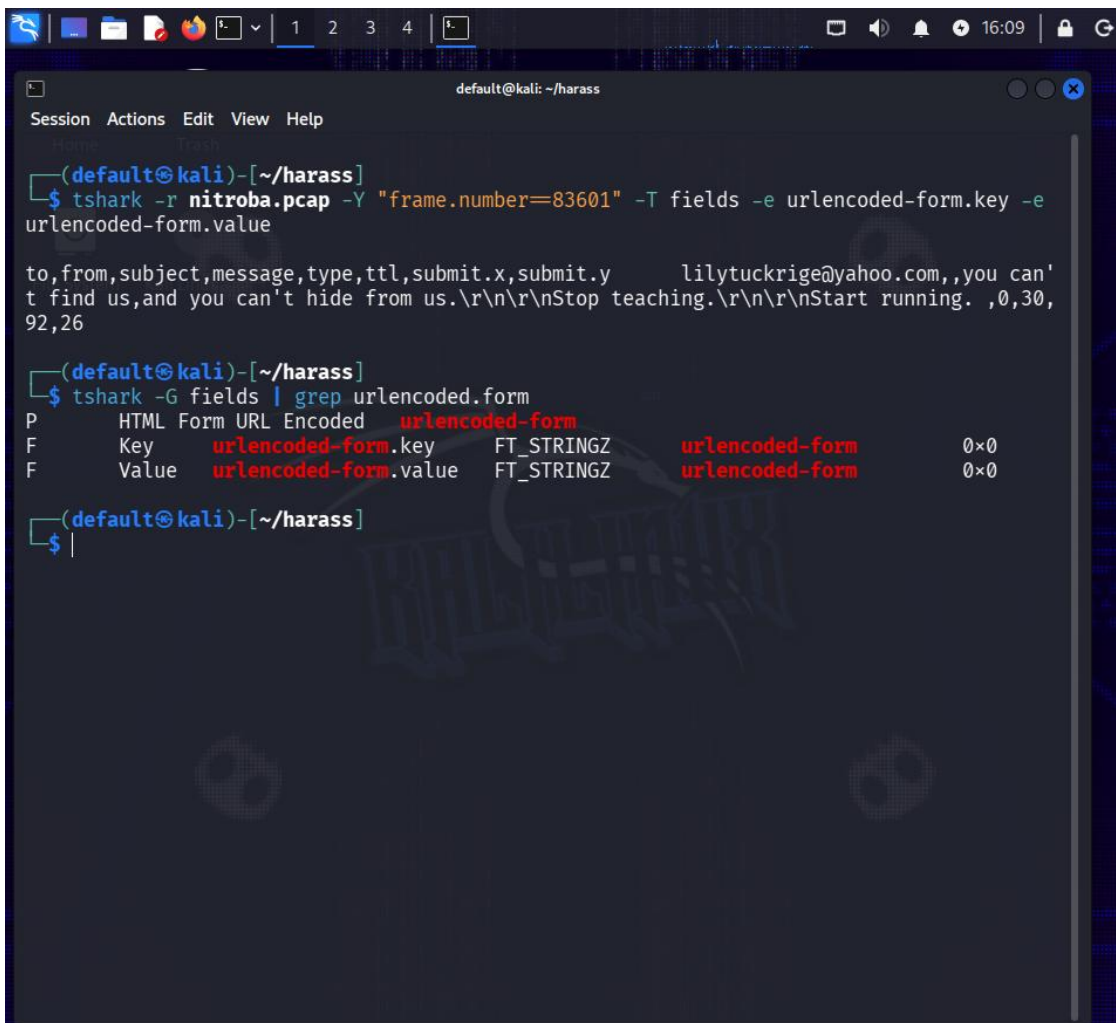
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y \"frame.number == 83601\" -V | grep -i \"form item\" --color

Form item: \"to\" = \"lilytuckrige@yahoo.com\"
Form item: \"from\" = \"\"
Form item: \"subject\" = \"you can't find us\"
Form item: \"message\" = \"and you can't hide from us.\\r\\n\\r\\nStop teaching.\\r\\n\\r\\nStart running. \"
Form item: \"type\" = \"0\"
Form item: \"ttl\" = \"30\"
Form item: \"submit.x\" = \"92\"
Form item: \"submit.y\" = \"26\"

(default@kali)-[~/harass]
$

```

Fig. A-20 (16:06) — Clean listing of all nine submitted form fields (frame 83601).



```
default@kali: ~/harass
Session Actions Edit View Help
Home Trash
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "frame.number==83601" -T fields -e urlencoded-form.key -e urlencoded-form.value

to,from,subject,message,type,ttl,submit.x,submit.y      lilytuckrige@yahoo.com,,you can'
t find us,and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running. ,0,30,
92,26

(default@kali)-[~/harass]
$ tshark -G fields | grep urlencoded.form
P      HTML Form URL Encoded  urlencoded-form
F      Key      urlencoded-form.key      FT_STRINGZ      urlencoded-form      0x0
F      Value    urlencoded-form.value      FT_STRINGZ      urlencoded-form      0x0

(default@kali)-[~/harass]
$ |
```

Fig. A-21 (16:09) — Confirmation of the urlencoded-form.key/value TShark field definitions.


```

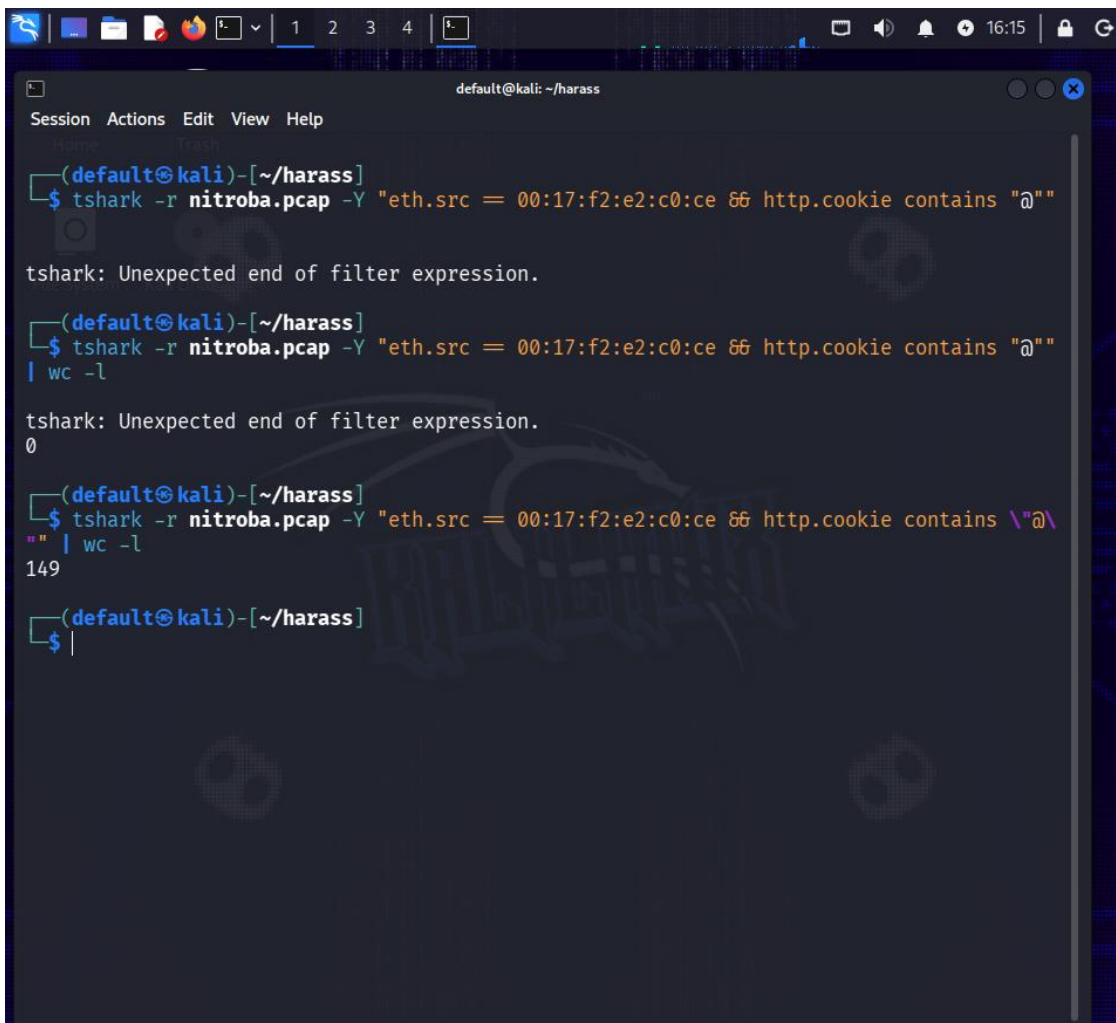
default@kali: ~/harass
Session Actions Edit View Help
Home Trash
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "frame.number == 83601" -T fields -e eth.src
00:17:f2:e2:c0:ce
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "eth.addr == 00:17:f2:e2:c0:ce" | wc -l
73246
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -qz io,stat,0,"COUNT(frame) frame","eth.addr == 00:17:f2:e2:c0:ce"

=====
| IO Statistics                                     |
|-----|-----|
| Duration: 15760.0 secs                         |
| Interval: 15760.0 secs                        |
| Col  1: COUNT(frame) frame                     |
|      2: eth.addr == 00:17:f2:e2:c0:ce          |
|-----|-----|
| Interval | 1 | 2 |          |
|           | COUNT | Frames | Bytes |
|-----|-----|
| 0.0 < 15760.0 | 94410 | 73246 | 44742268 |
|-----|-----|

(default@kali)-[~/harass]
$

```

Fig. A-22 (16:13) — *eth.src* confirmed for frame 83601; *io,stat* quantifies MAC traffic volume.



The screenshot shows a Kali Linux terminal window with the title bar "default@kali: ~/harass". The terminal displays the following commands and output:

```
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@""

tshark: Unexpected end of filter expression.

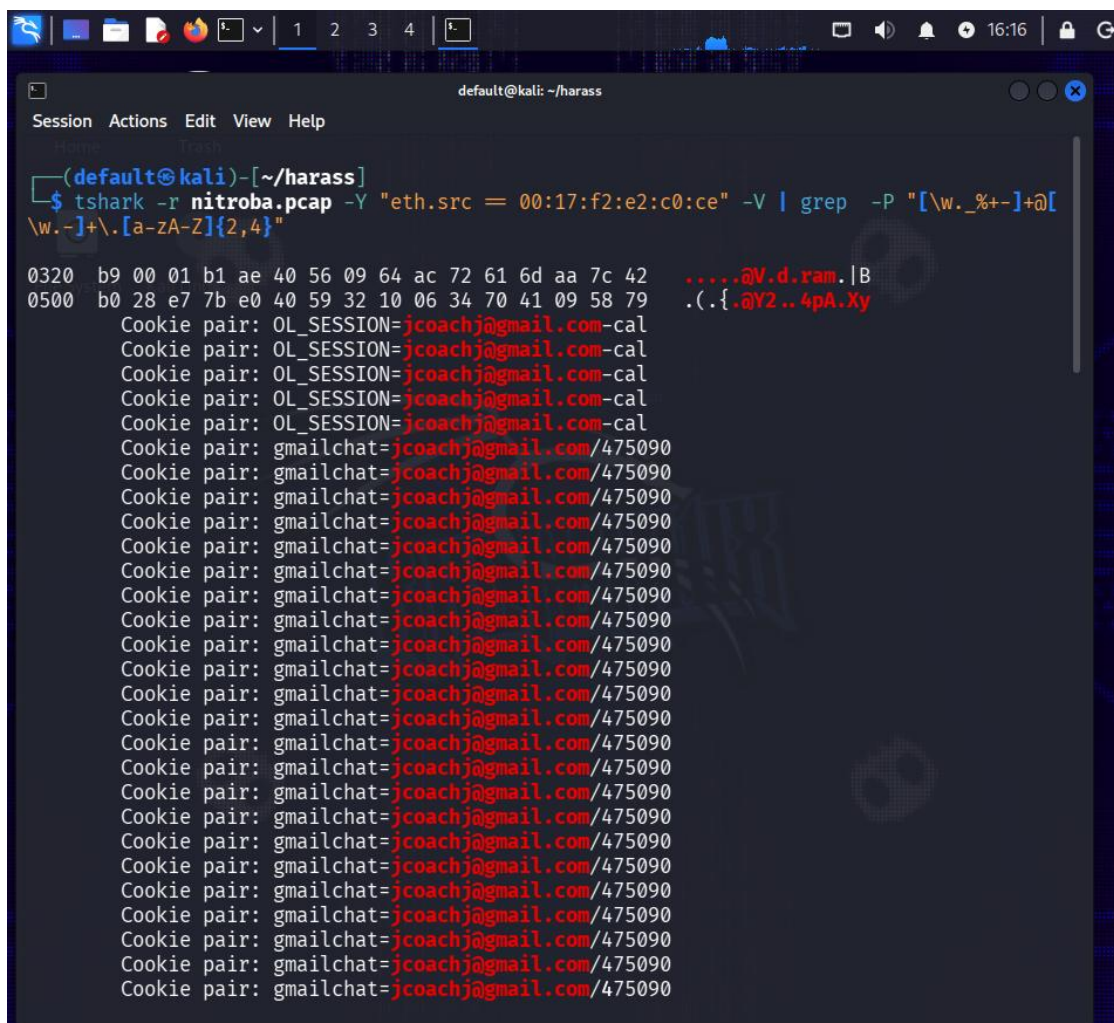
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "eth.src == 00:17:f2:e2:c0:ce && http.cookie contains @"
| wc -l

tshark: Unexpected end of filter expression.
0

(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "eth.src == 00:17:f2:e2:c0:ce && http.cookie contains \"@\"
\" | wc -l
149

(default@kali)-[~/harass]
$ |
```

Fig. A-23 (16:15) — Count of "@"-bearing cookie frames from the client MAC (149 frames).



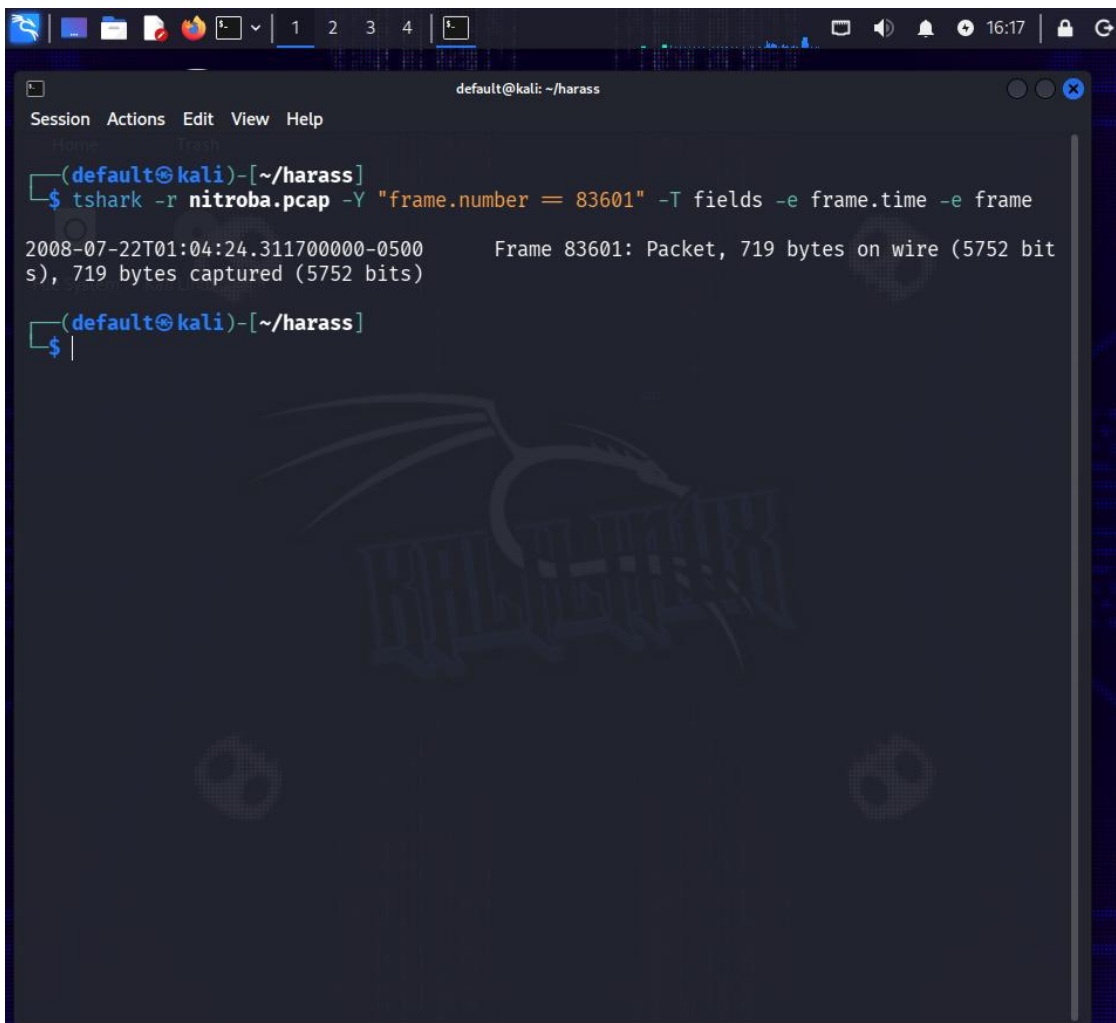
```

default@kali: ~/harass
Session Actions Edit View Help
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "eth.src == 00:17:f2:e2:c0:ce" -V | grep -P "[\w._%+-]+\@[\w.-]+\.[a-zA-Z]{2,4}"

0320  b9 00 01 b1 ae 40 56 09 64 ac 72 61 6d aa 7c 42      .....@V.d.ram.|B
0500  b0 28 e7 7b e0 40 59 32 10 06 34 70 41 09 58 79      .(.{. @Y2 .. 4pA.Xy
Cookie pair: OL_SESSION=jcoachj@gmail.com-cal
Cookie pair: OL_SESSION=jcoachj@gmail.com-cal
Cookie pair: OL_SESSION=jcoachj@gmail.com-cal
Cookie pair: OL_SESSION=jcoachj@gmail.com-cal
Cookie pair: OL_SESSION=jcoachj@gmail.com-cal
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090
Cookie pair: gmailchat=jcoachj@gmail.com/475090

```

Fig. A-24 (16:16) — Regex extraction enumerates every recovered e-mail-pattern cookie value.



The screenshot shows a Kali Linux terminal window with the title bar 'default@kali: ~/harass'. The terminal has a menu bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The user is at the prompt '(default@kali)-[~/harass]'. They enter the command: `$ tshark -r nitroba.pcap -Y "frame.number == 83601" -T fields -e frame.time -e frame`. The output shows the timestamp '2008-07-22T01:04:24.311700000-0500' and the frame details 'Frame 83601: Packet, 719 bytes on wire (5752 bits), 719 bytes captured (5752 bits)'. The terminal background has a faint 'RELENTLESS' watermark.

```
(default@kali)-[~/harass]
$ tshark -r nitroba.pcap -Y "frame.number == 83601" -T fields -e frame.time -e frame
2008-07-22T01:04:24.311700000-0500      Frame 83601: Packet, 719 bytes on wire (5752 bit
s), 719 bytes captured (5752 bits)
(default@kali)-[~/harass]
$
```

Fig. A-25 (16:17) — Exact POST frame timestamp re-confirmed for the integrated timeline.

2. Chem 109 - Class roster

Teacher: Lily Tuckrige

Students:

Amy Smith
Burt Greedom
Tuck Gorge
Ava Book
Johnny Coach
Jeremy Ledvkin
Nancy Colburne
Tamara Perkins
Esther Pringle
Asar Misrad
Jenny Kant

Fig. A-26 — Supplied Chemistry 109 class roster, used for the comparison in §4.6.